
Indefinite causal order in boxworld theories

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arXiv > quant-ph > arXiv:2411.00951

Quantum Physics

[Submitted on 1 Nov 2024]

Indefinite causal order in boxworld theories

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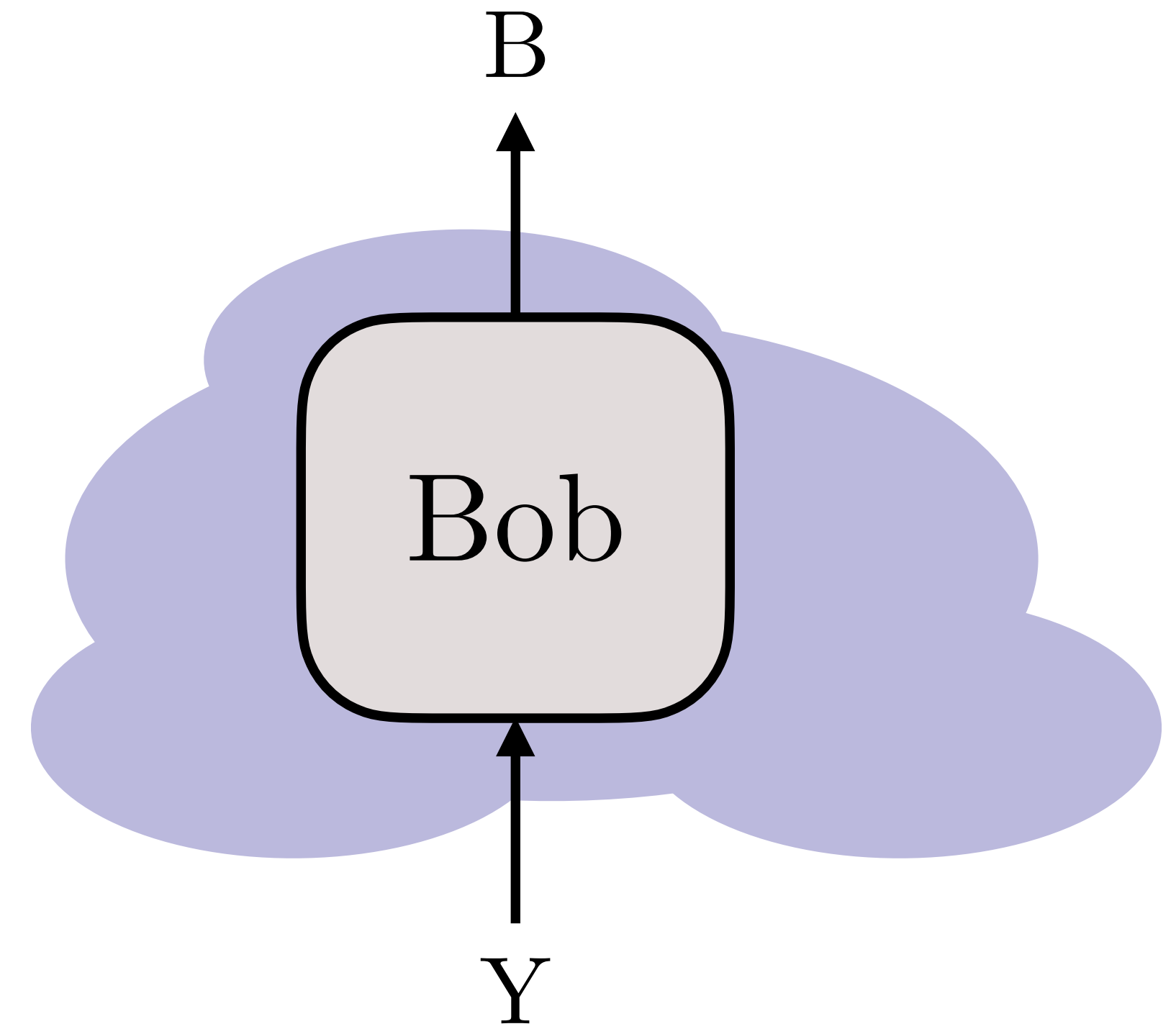
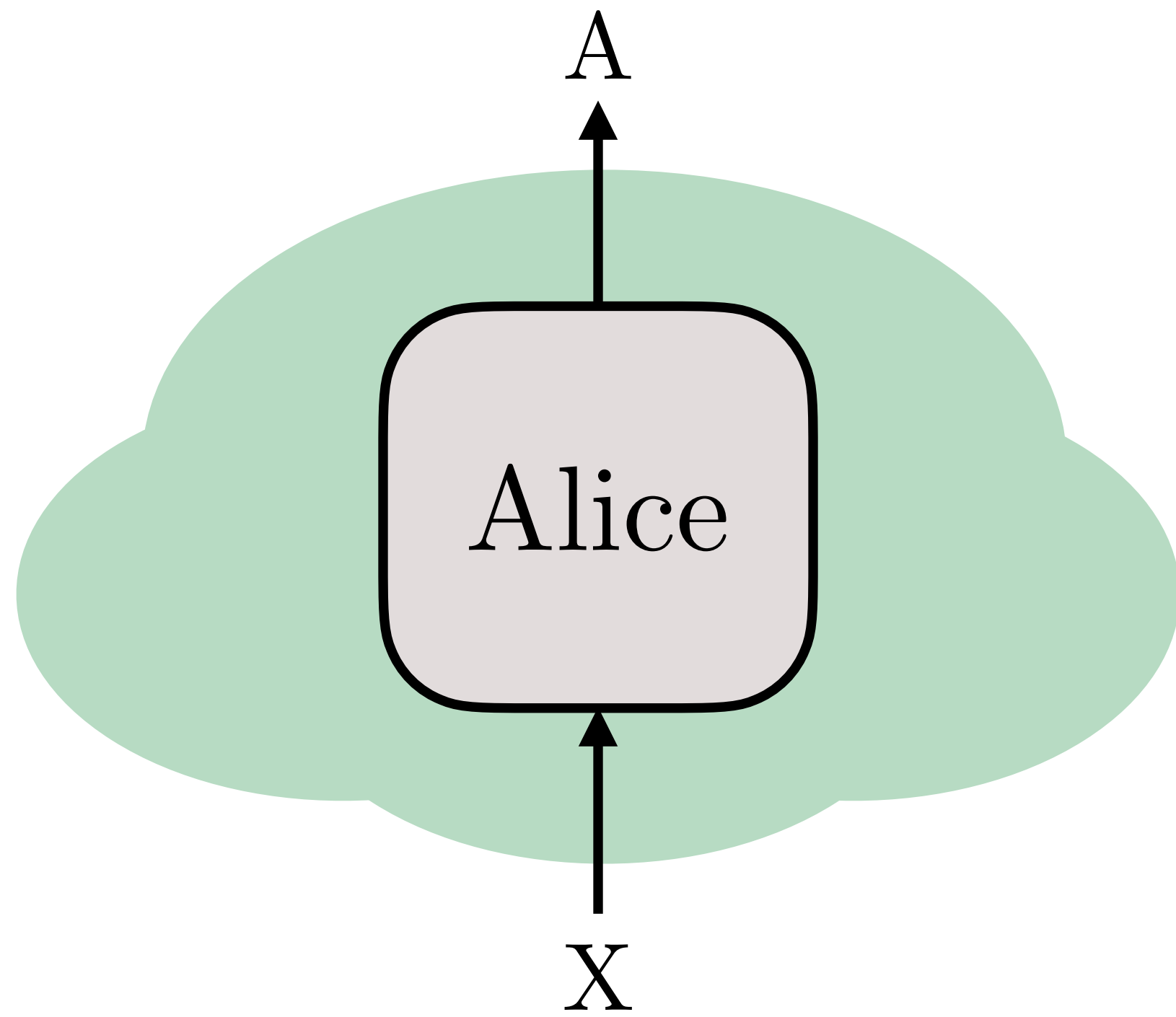
Correlations

This talk is about probability distributions.

$$P_{AB|XY}$$

correlations

Correlations



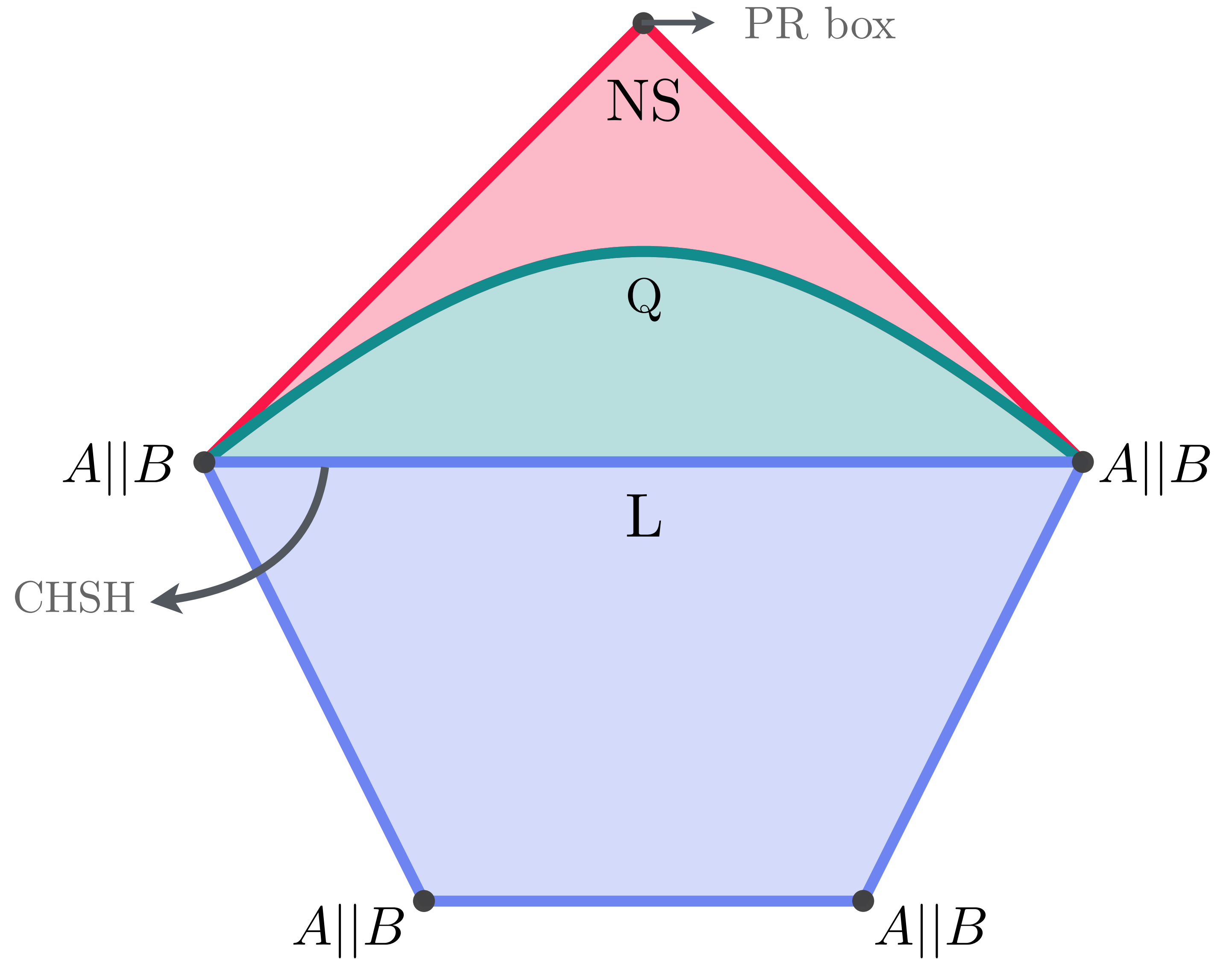
Independent labs collect sets of joint conditional probability distributions

$$P_{AB|XY}$$

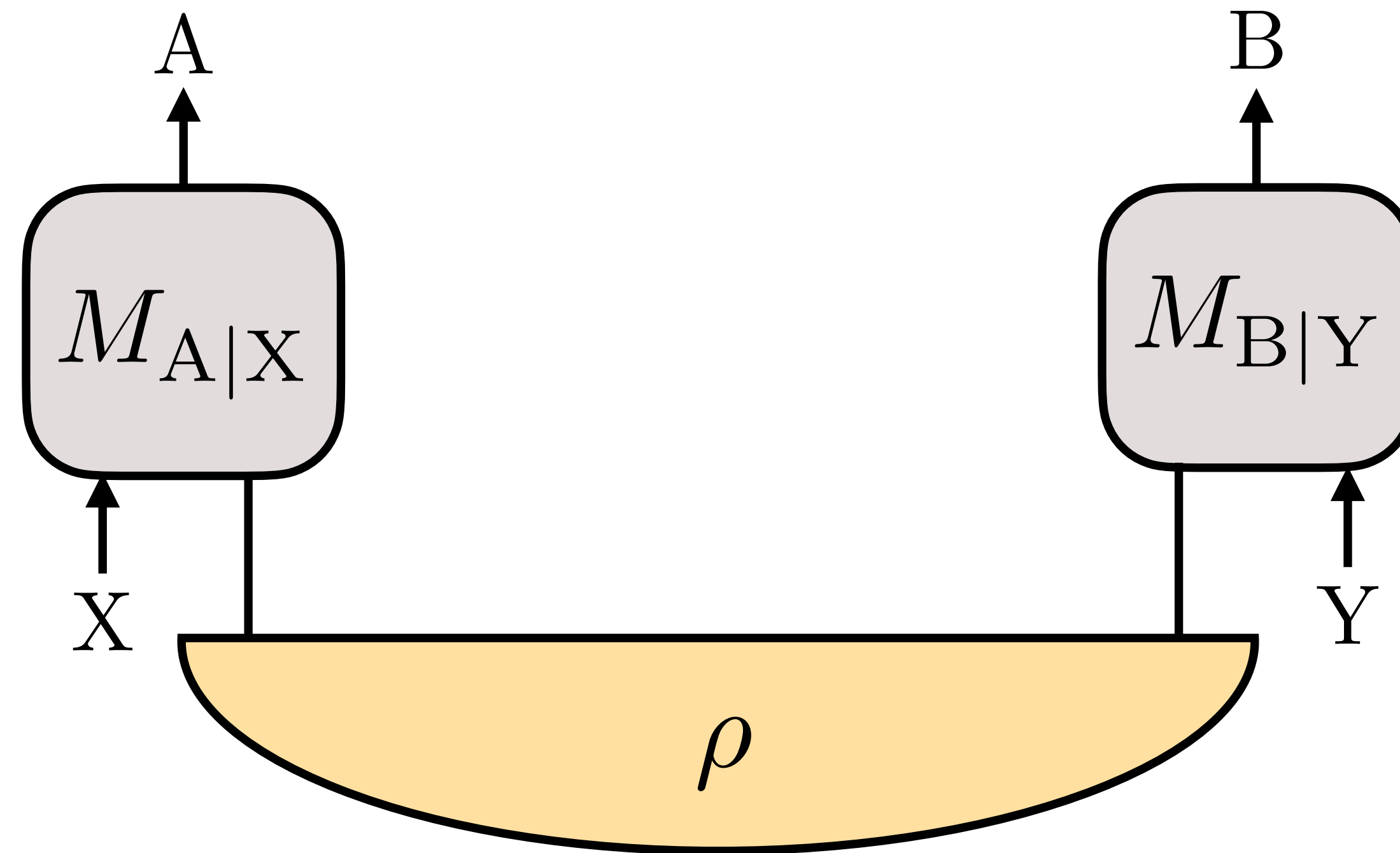
Nonsignaling correlations

$$\sum_B P_{AB|XY}^{\text{NS}} = P_{A|X}$$

$$\sum_A P_{AB|XY}^{\text{NS}} = P_{B|Y}$$



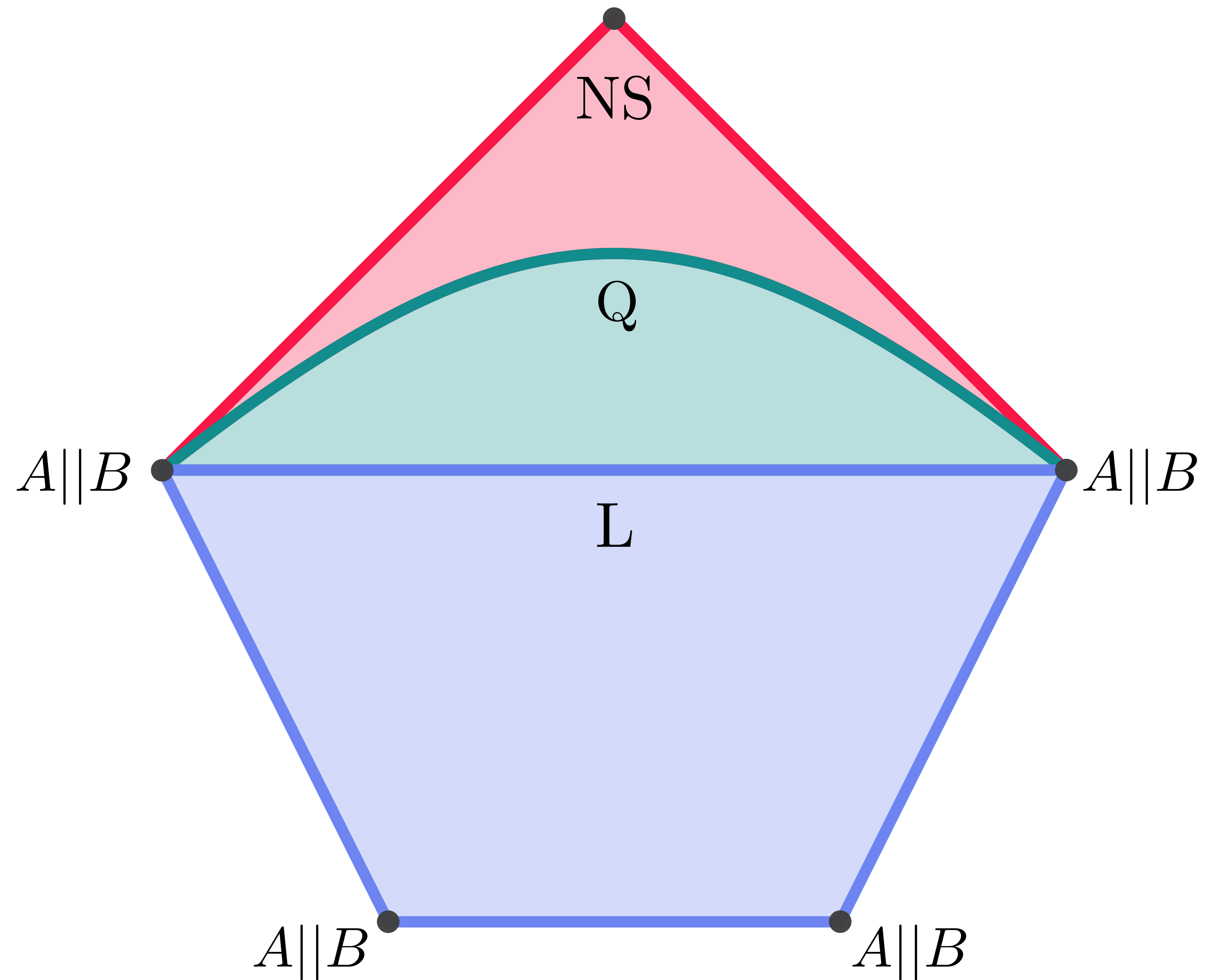
Nonsignaling quantum correlations



$$P_{AB|XY} = \text{tr}[(M_{A|X} \otimes M_{B|Y}) \rho]$$

Created by quantum systems and measurements in Bell scenarios

Nonsignaling correlations



Nonsignaling correlations

Physical principles to delimitate the set of NS quantum correlations:

- **Nontrivial communication complexity**

v. Dam, arxiv:quant-ph/0501159

Brassard et al., PRL (2006)

- **Information causality**

Pawłowski et al., Nature (2009)

- **Macroscopic locality**

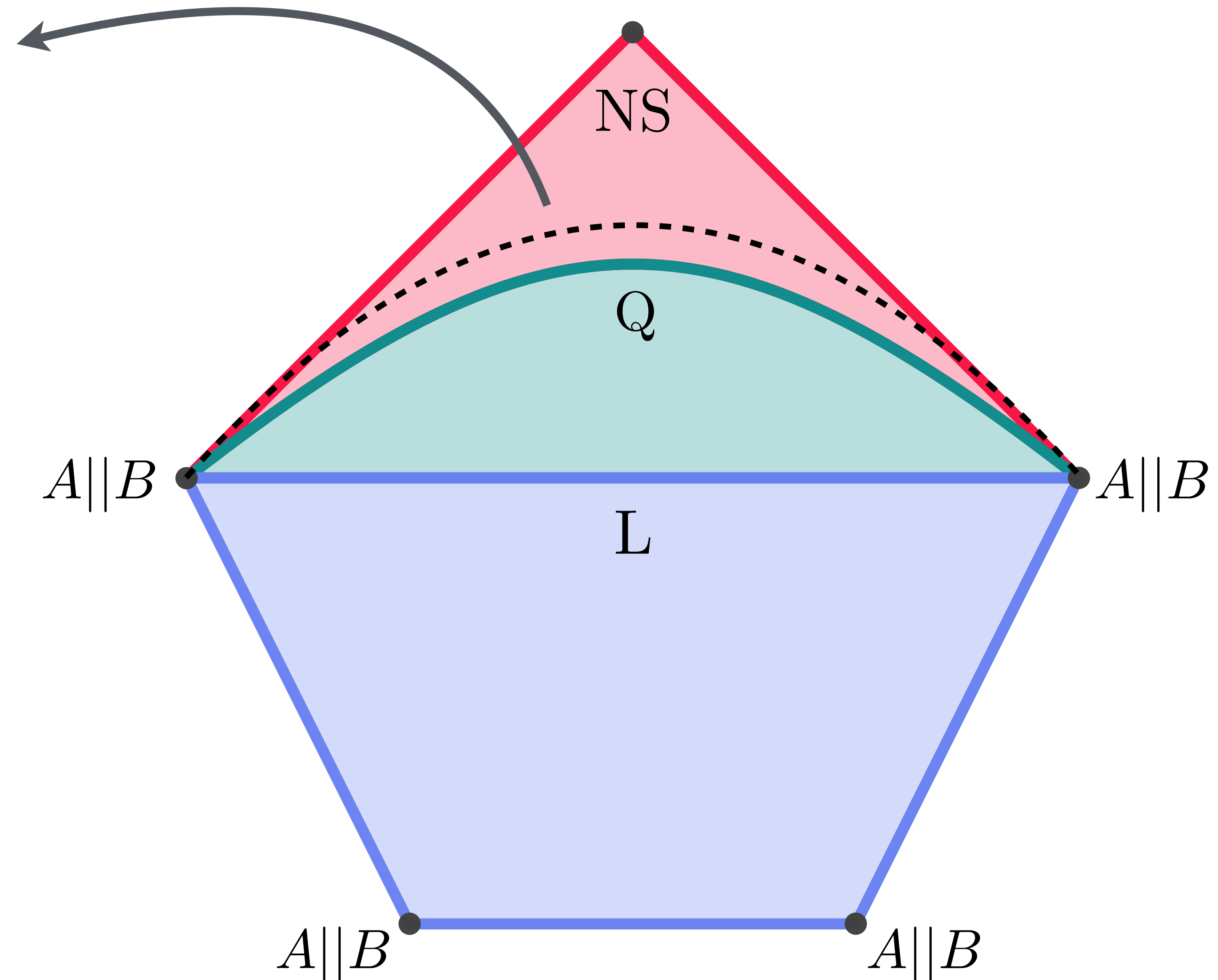
Navascués et al., PRSA (2009)

- **Local orthogonality**

Fritz et al., Nat. Commun (2013)

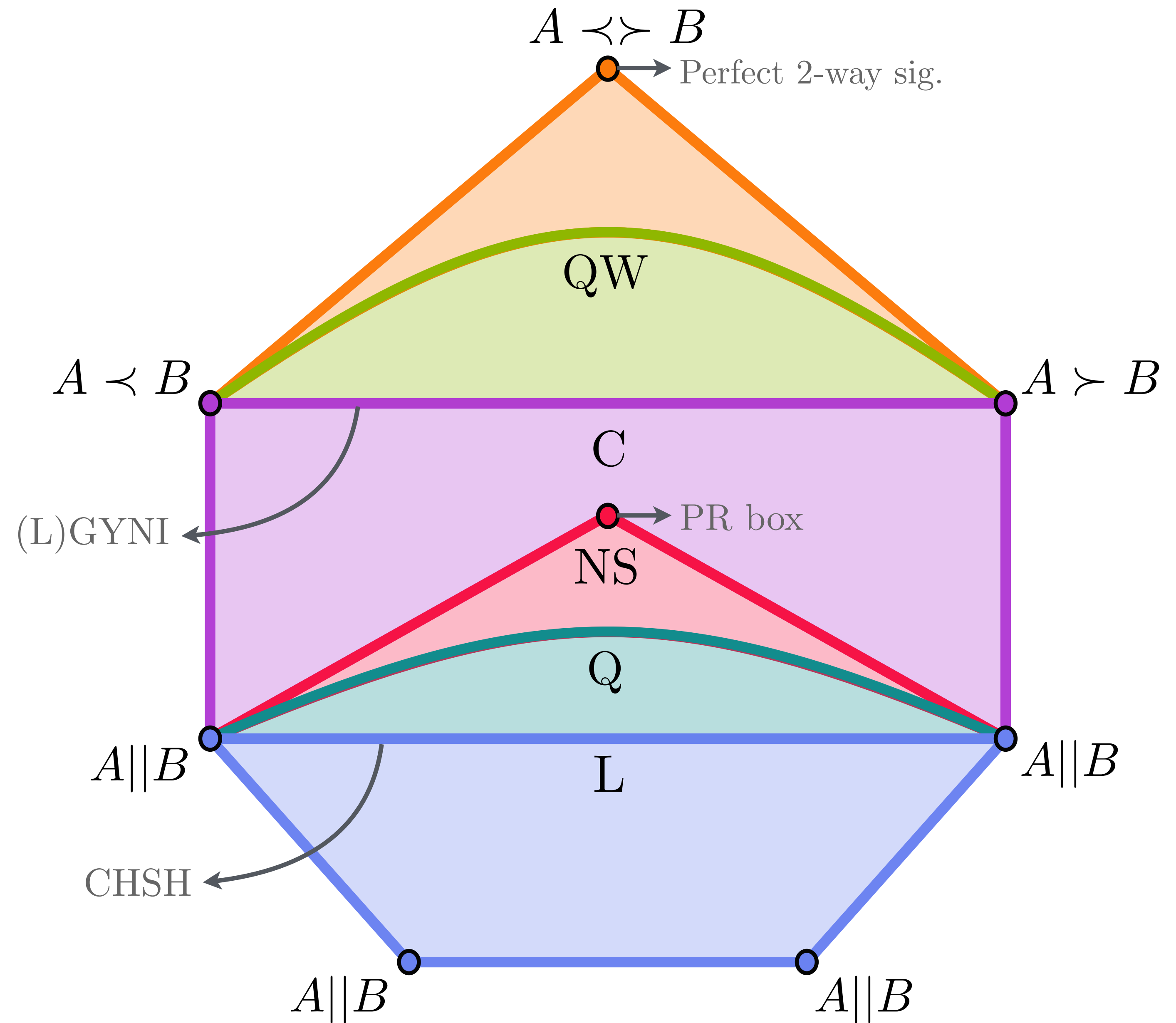
- **E-principle**

Cabello, PRL (2013)



All correlations

$$P_{AB|XY}$$

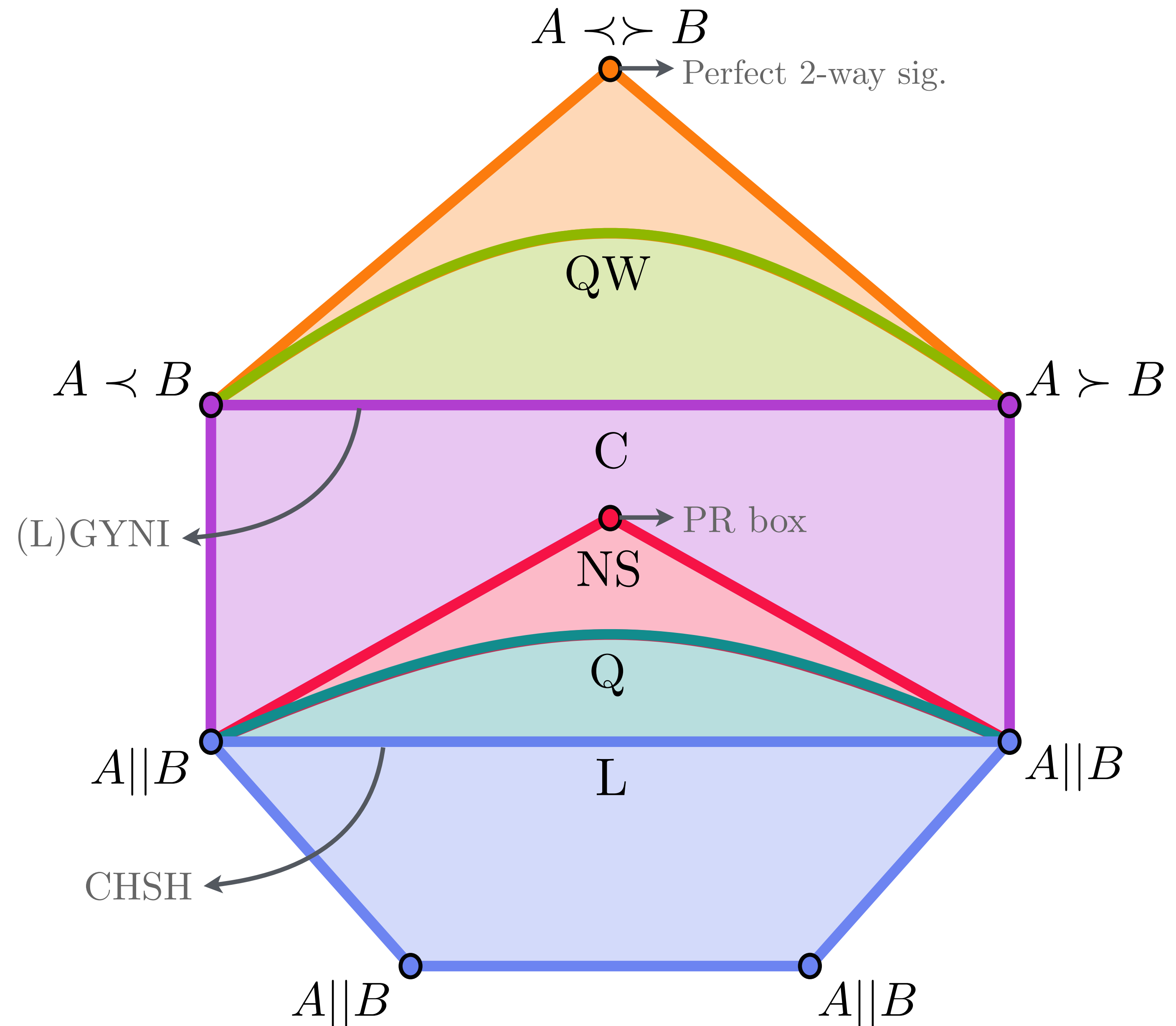


All correlations

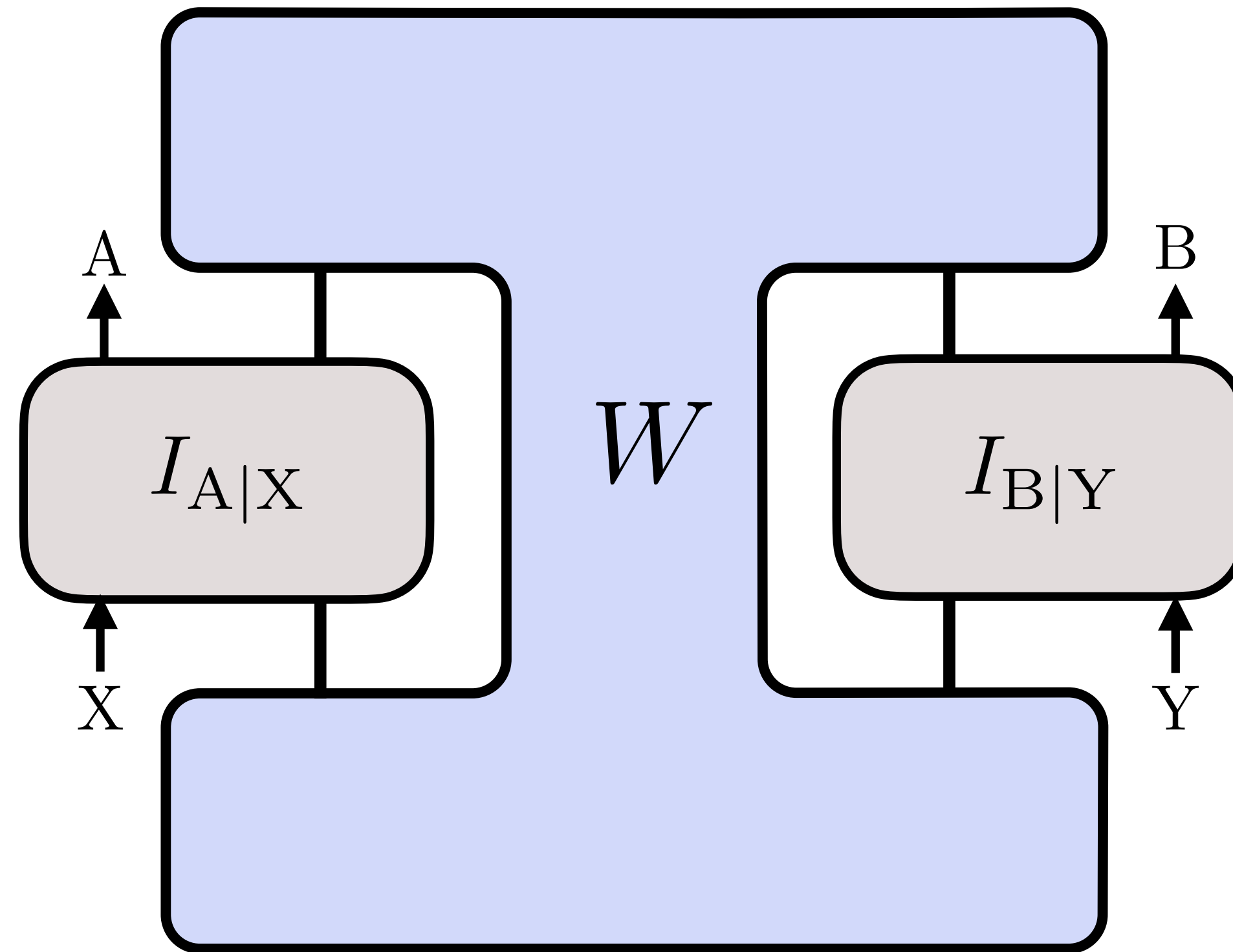
$$P_{AB|XY}$$

$$P_{AB|XY}^{\text{causal}} = q P_{AB|XY}^{A \prec B} + (1 - q) P_{AB|XY}^{B \prec A}$$

Genuine two-way signalling correlations
are called noncausal correlations



Process matrix correlations

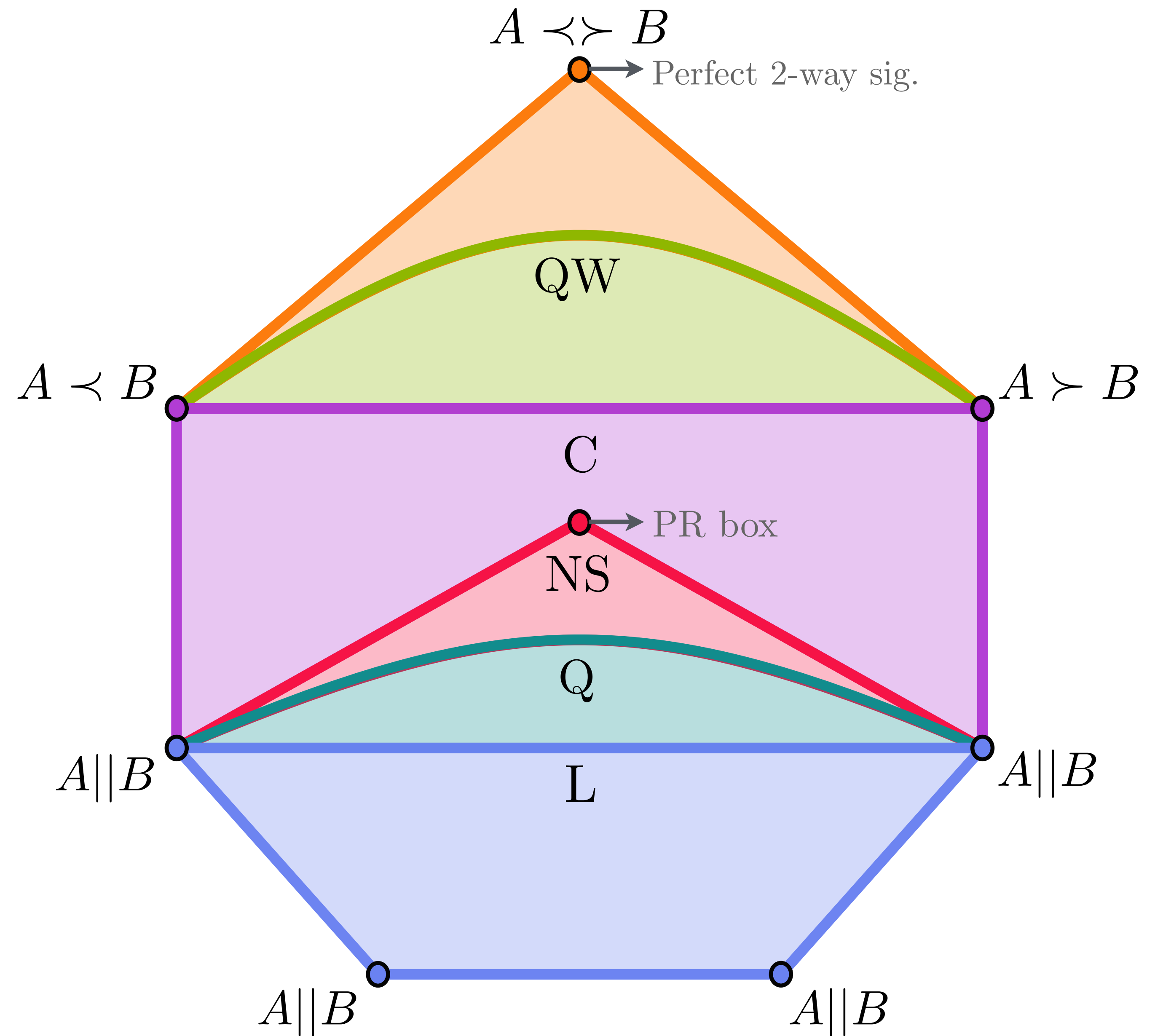


$$P_{AB|XY} = \text{tr}[(I_{A|X} \otimes I_{B|Y}) W]$$

Created by process matrices and quantum instruments in higher-order quantum theory

All correlations

Process matrix correlations



All correlations

Process matrix correlations

- **No perfect two-way sig. in finite dimensions**

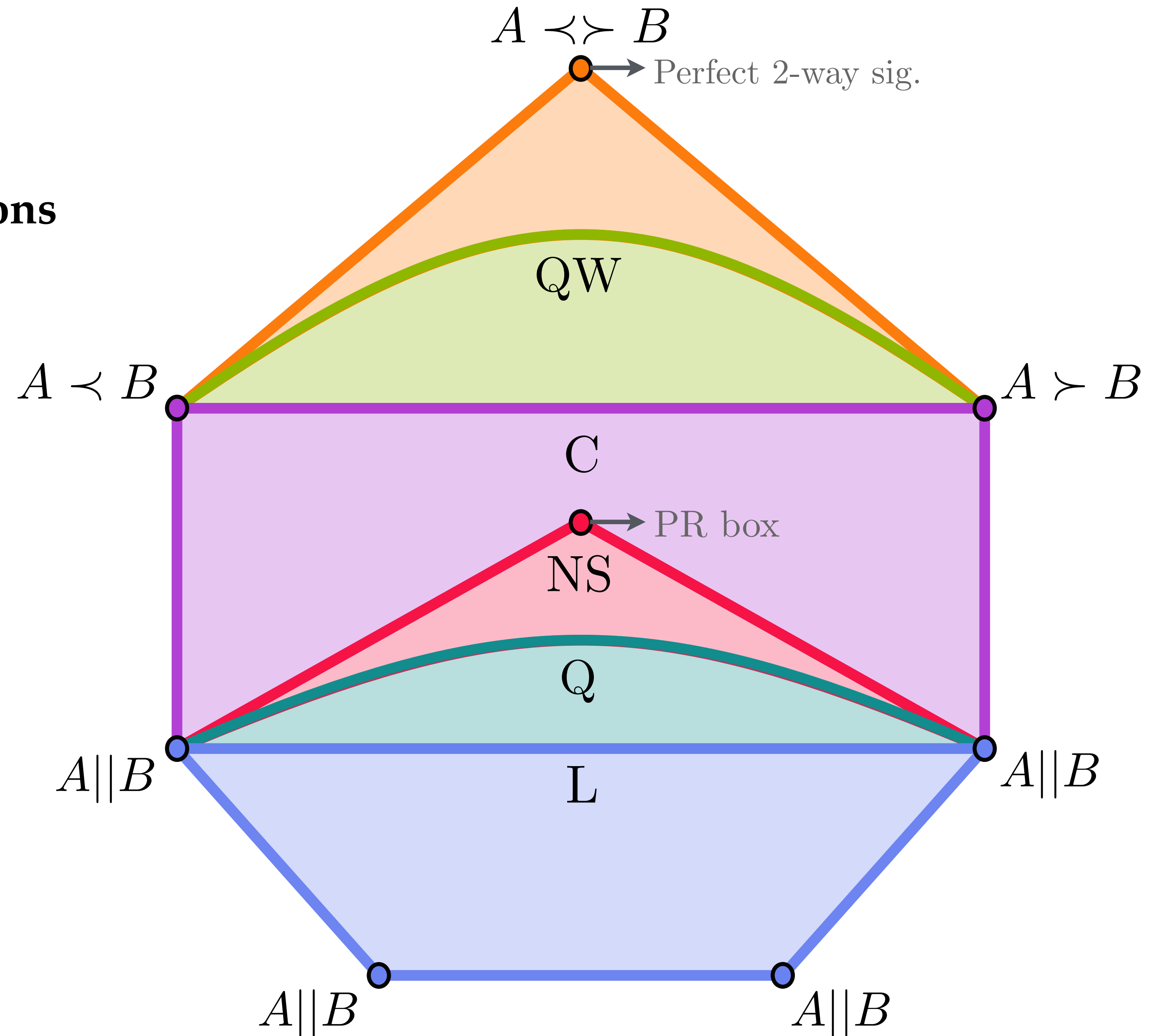
Bavaresco et al., Quantum (2019)

- **No perfect two-way sig. at all**

Kunjwal and Oreshkov, arXiv (2023)

- **Tsirelson-like bounds**

Liu and Chiribella, arXiv (2024)



All correlations

Process matrix correlations

- **No perfect two-way sig. in finite dimensions**

Bavaresco et al., Quantum (2019)

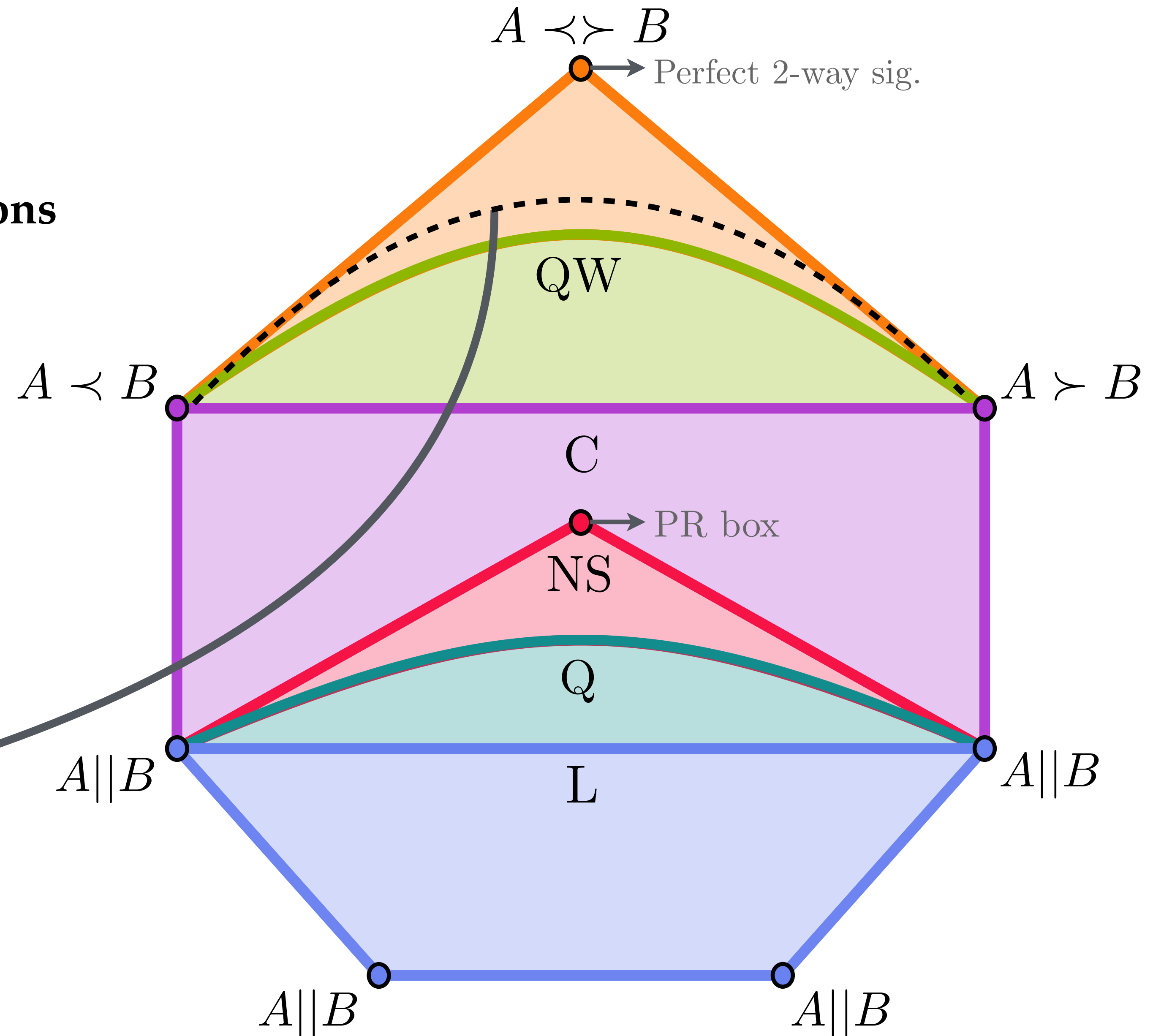
- No perfect two-way sig. at all

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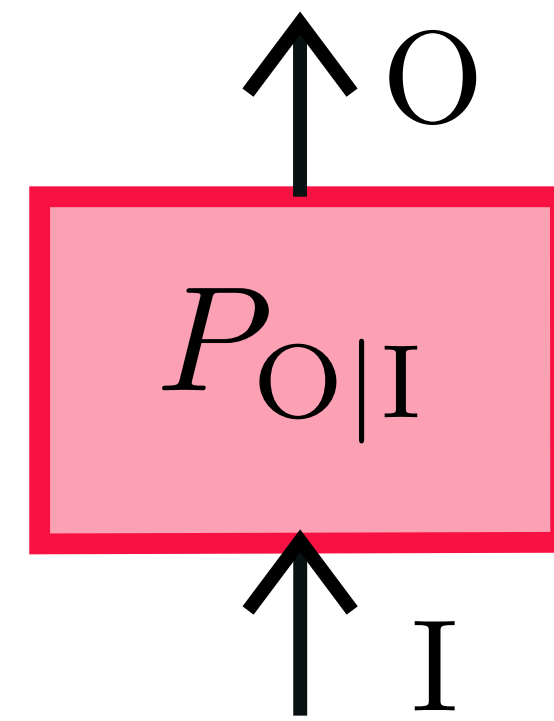
No outer approximation
given by physical principles



Boxworld

Boxworld

The generalized bit as the basic state of the theory

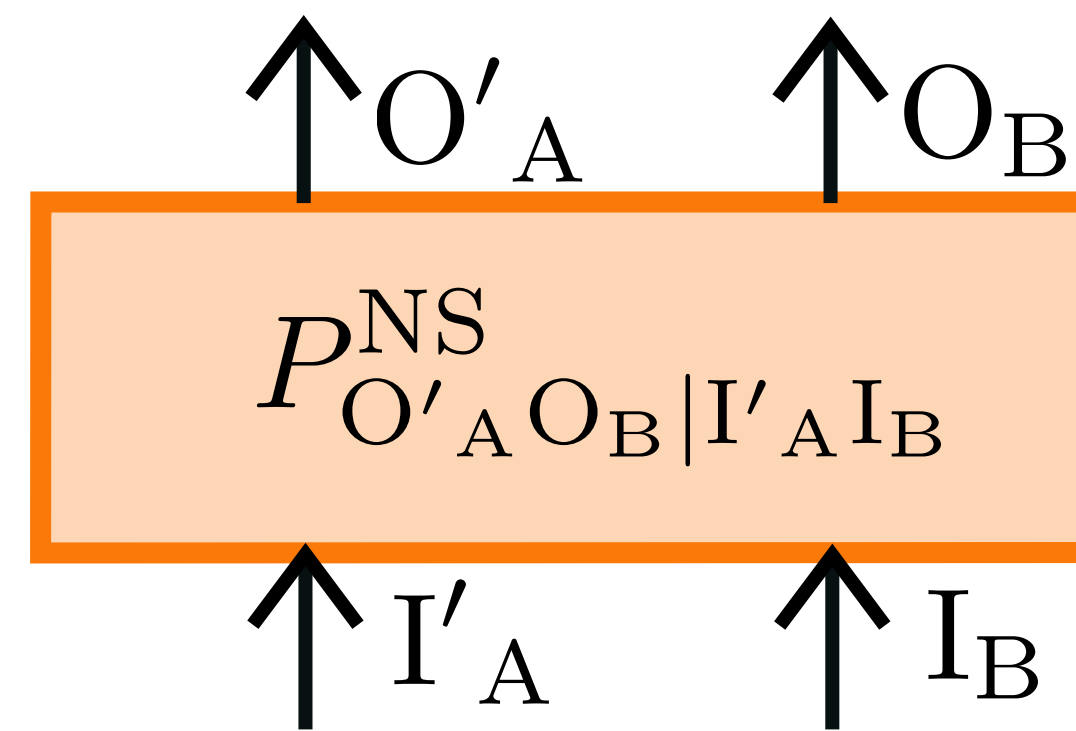


“box”

(a set of probability distributions)

Boxworld

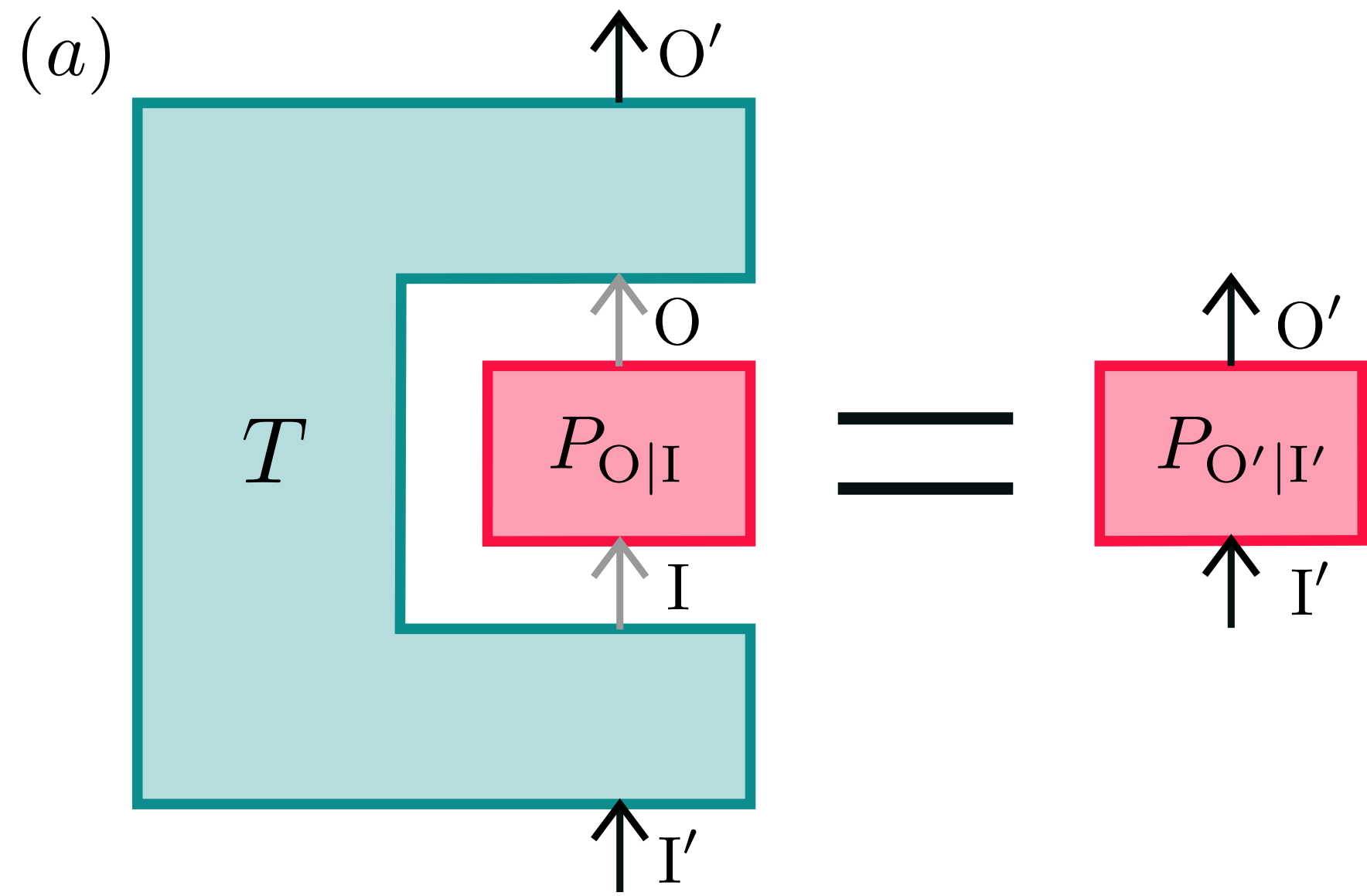
The generalized bit as the basic state of the theory



“bipartite box”

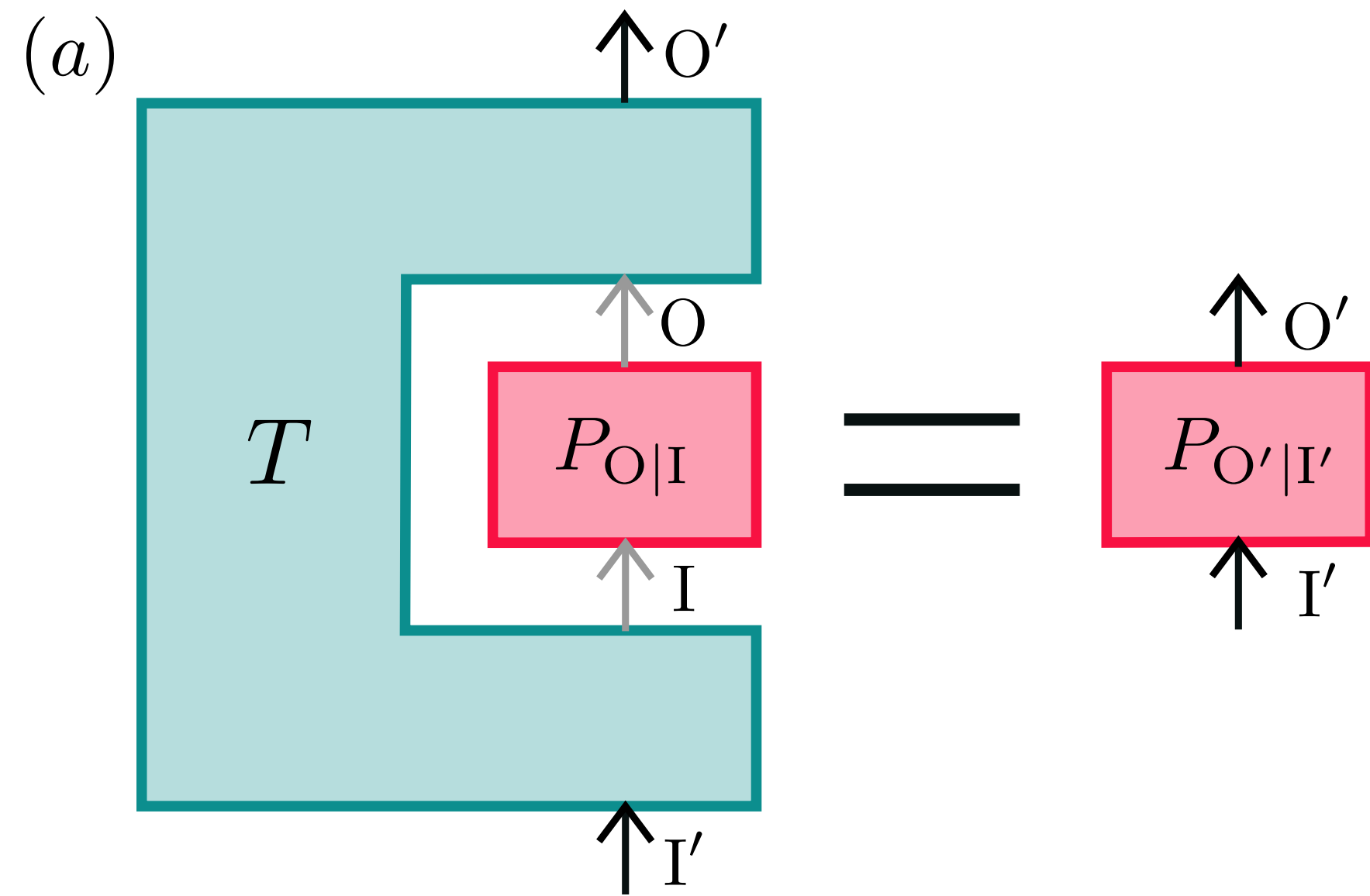
(a set of nonsignaling probability distributions)

Operations in boxworld

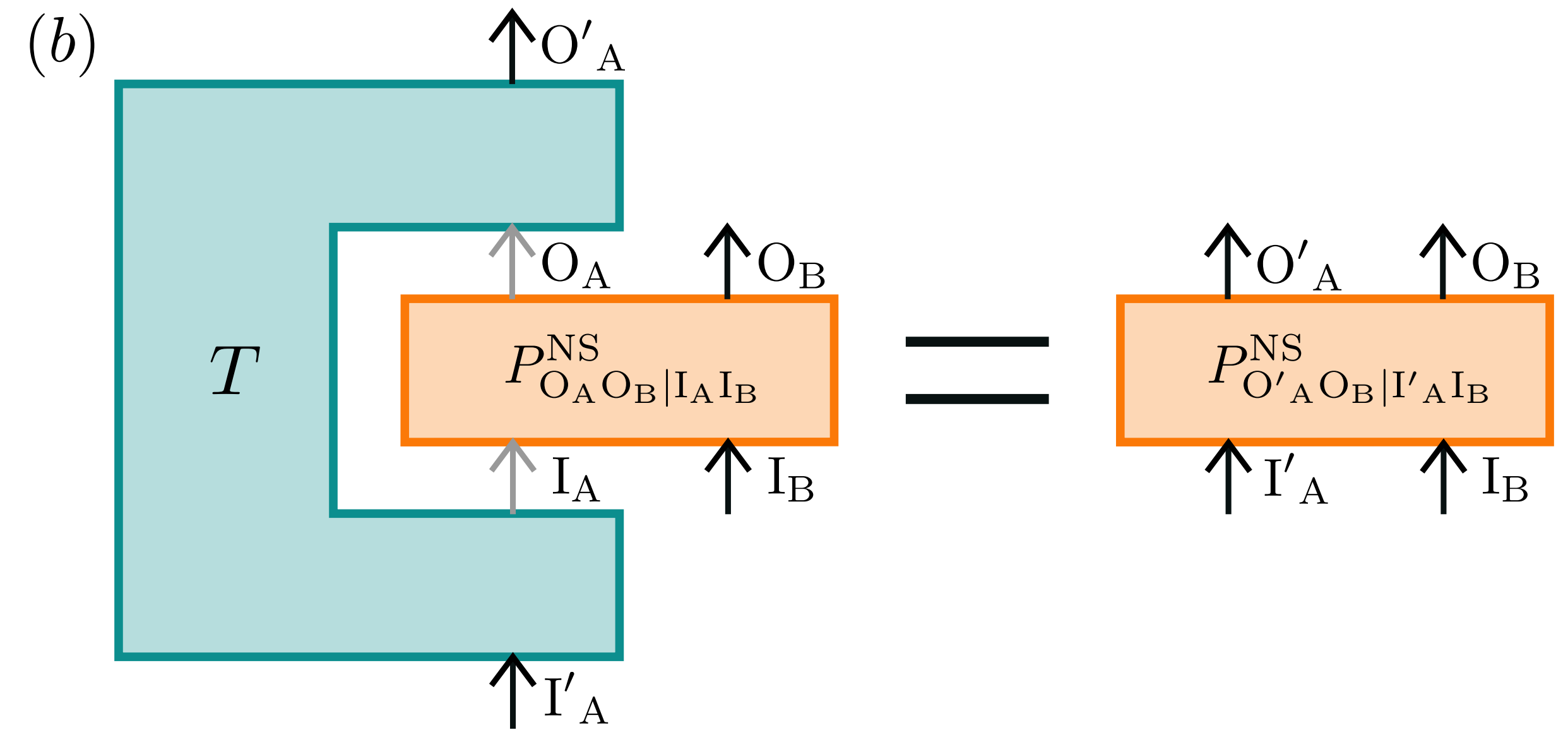


Maps boxes into boxes

Operations in boxworld

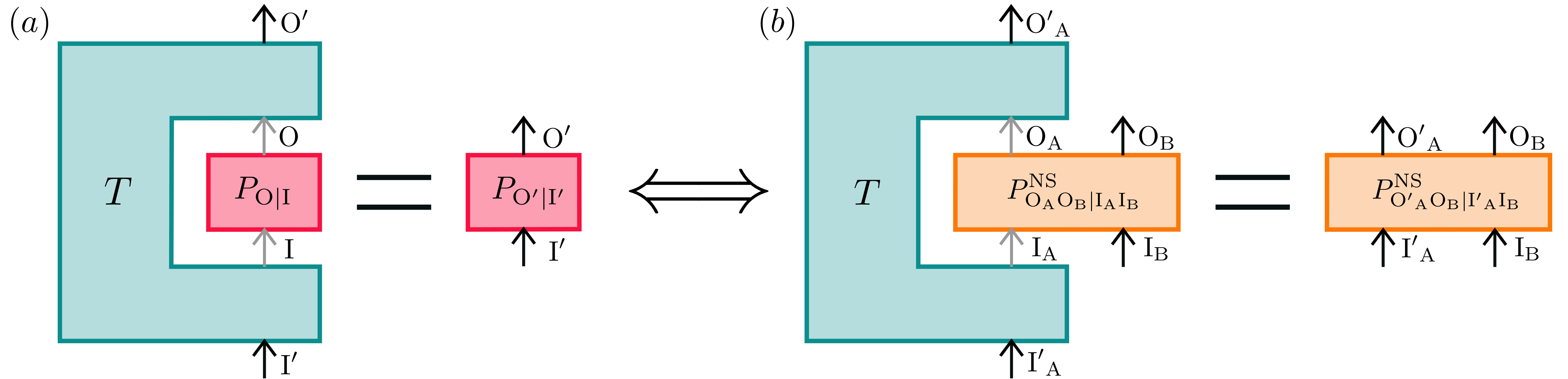


Maps boxes into boxes



Maps nonsignaling boxes into nonsignaling boxes
(when acting only on part of the box)

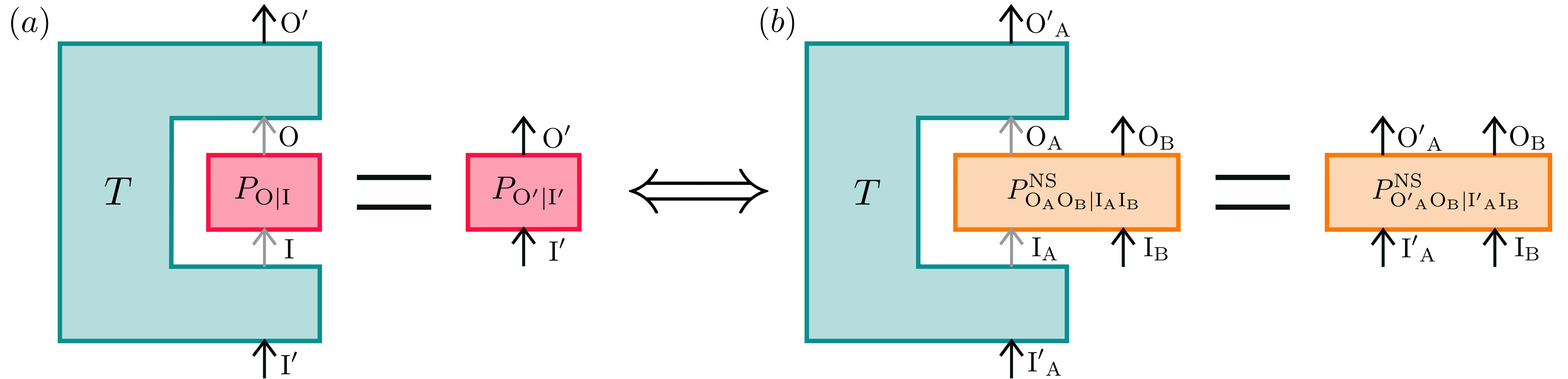
Operations in boxworld



Maps boxes into boxes

Maps nonsignaling boxes into nonsignaling boxes
(when acting only on part of the box)

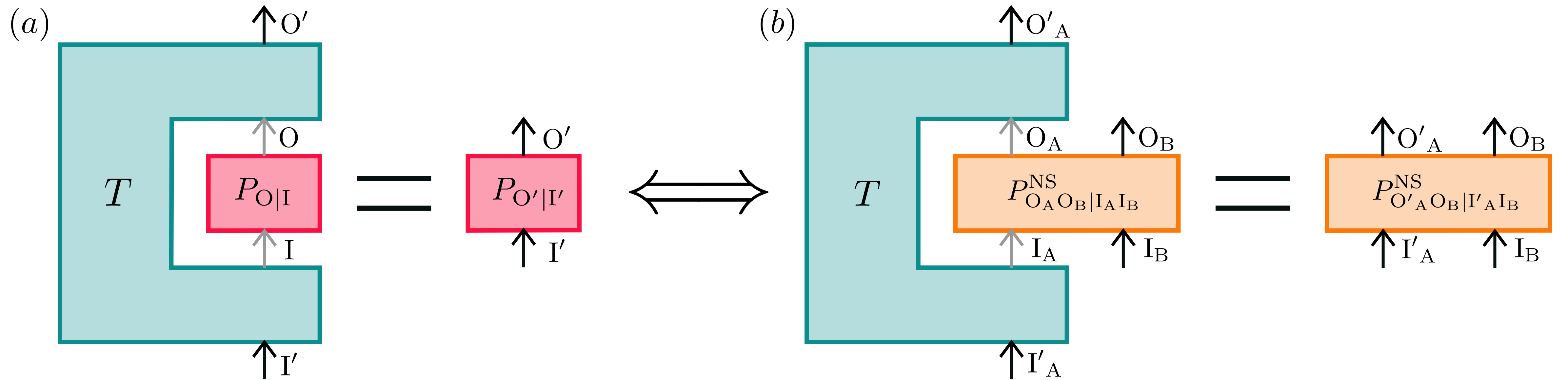
Operations in boxworld



$$T_{IO'|I'O} = P_{I|I'} P_{O'|I'IO} = \sum_{\lambda} \pi_{\lambda} D_{I|I'}^{\lambda} D_{O'|I'O}^{\lambda}$$

analog of quantum **channels**

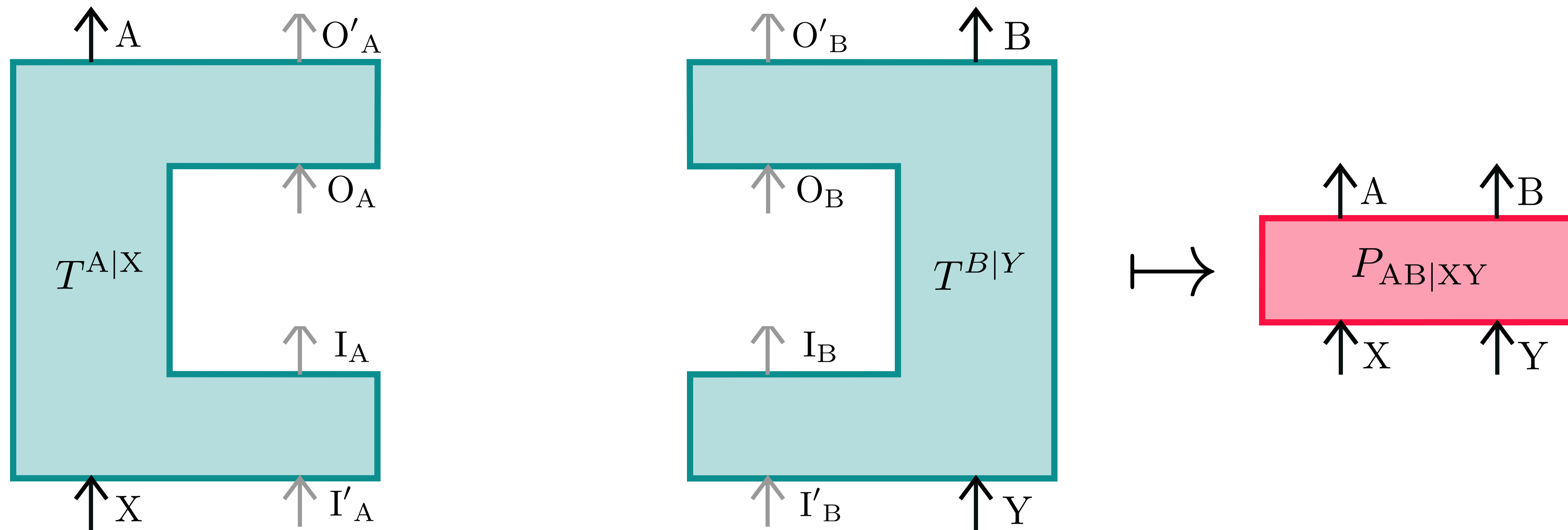
Operations in boxworld



$$T_{IO'|I'O}^{A|X} = \sum_{\lambda} \pi_{\lambda} D_{I|I'X}^{\lambda} D_{O'|I'OX}^{\lambda} D_{A|I'OX}^{\lambda}$$

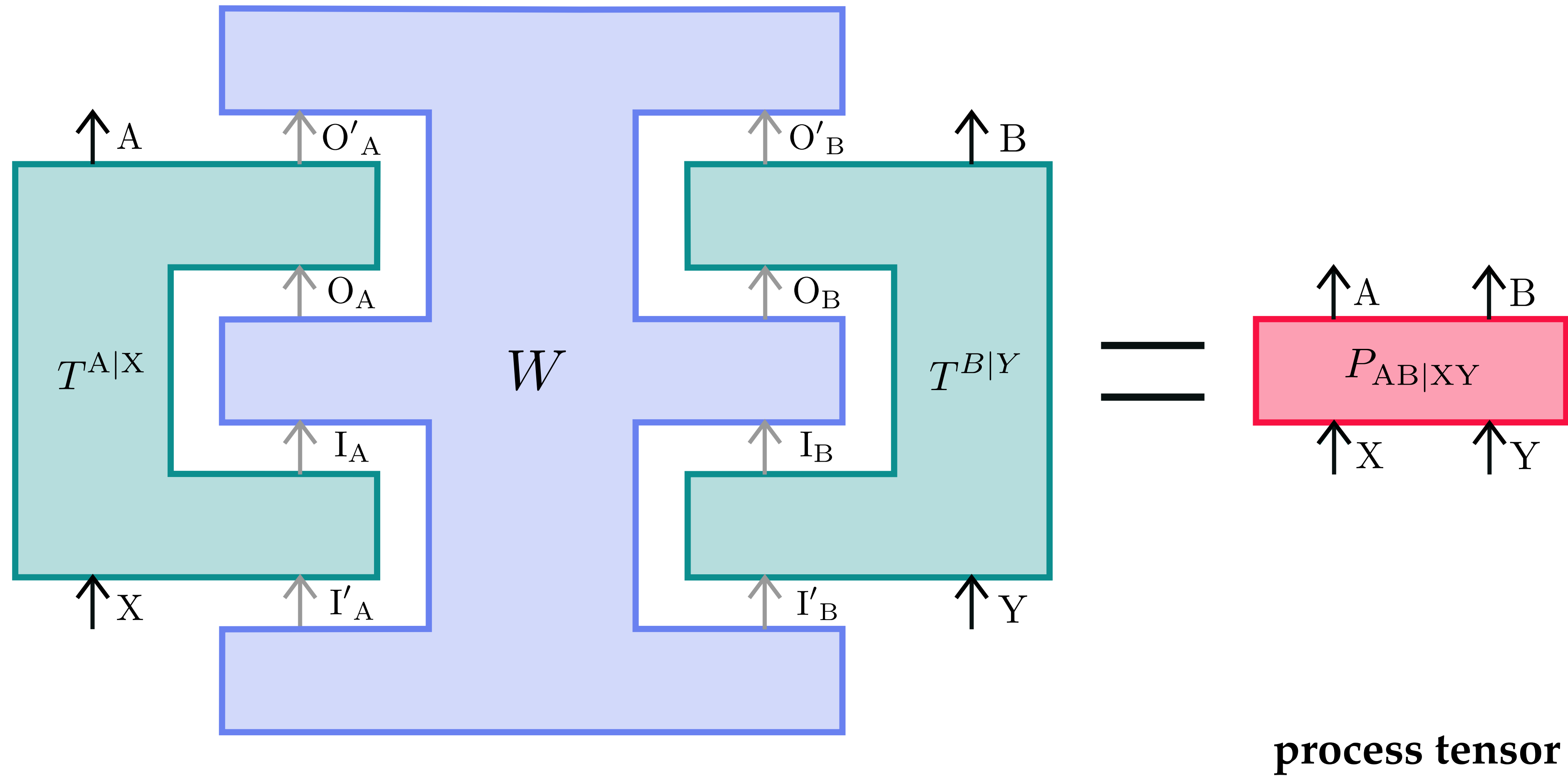
analog of quantum **instruments**

Higher-order boxworld theory

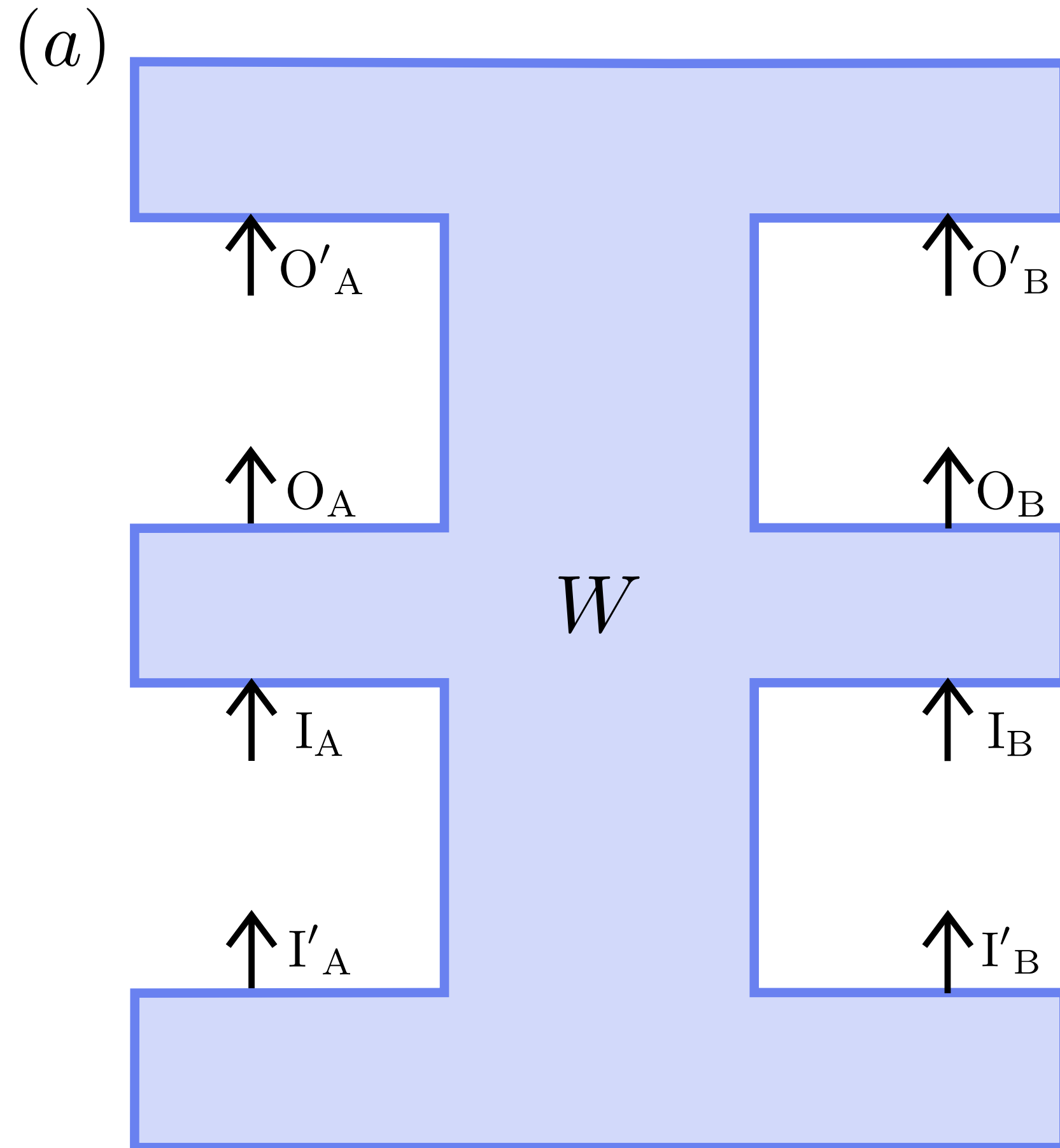


mapping probabilistic local operations to correlations
analog of a process matrix

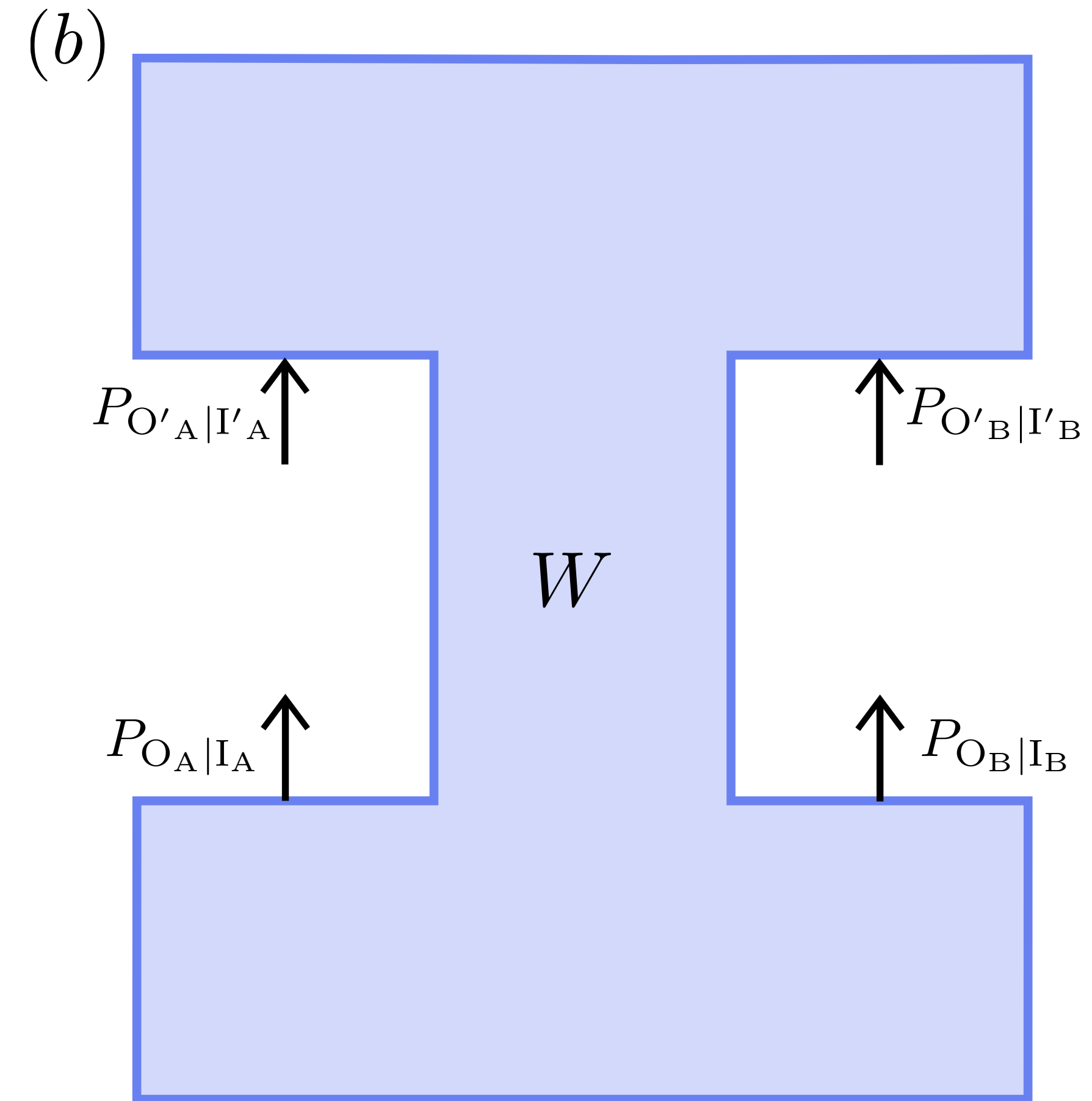
Higher-order boxworld theory



Process tensor representations



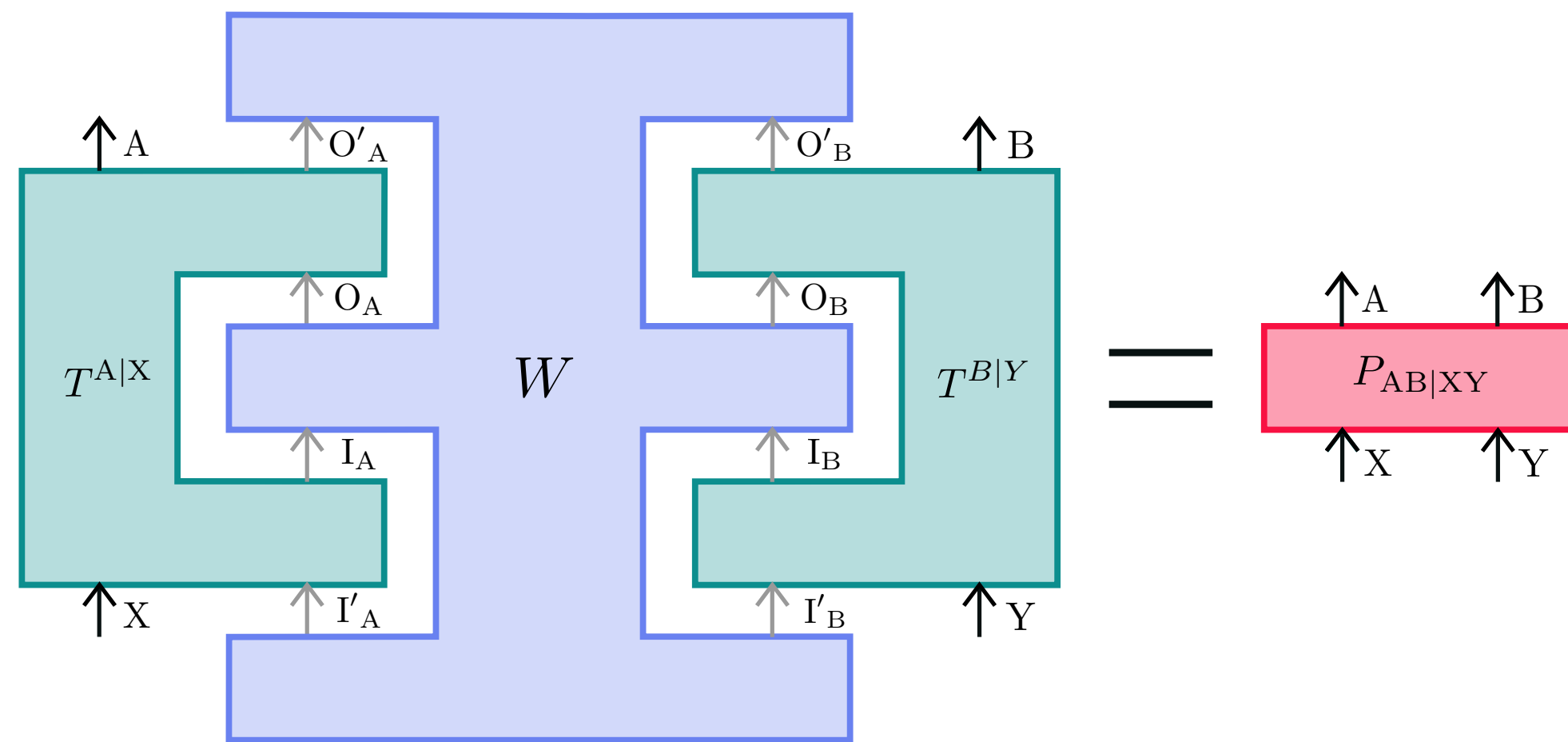
wires represent random variables



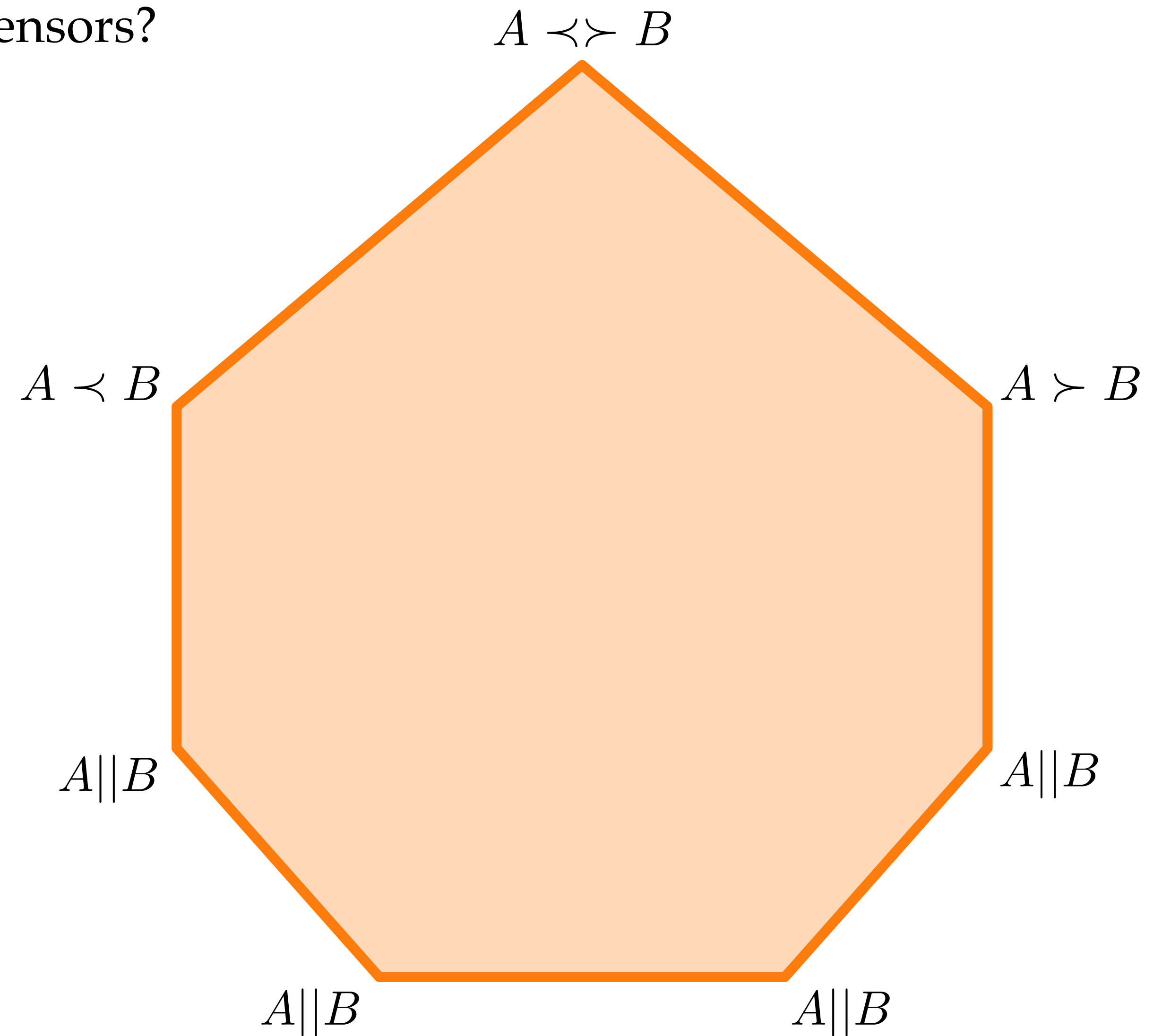
wires represent local systems (boxes)

Correlations from process tensors

What are the correlations that can be produced by local probabilistic operations acting on process tensors?

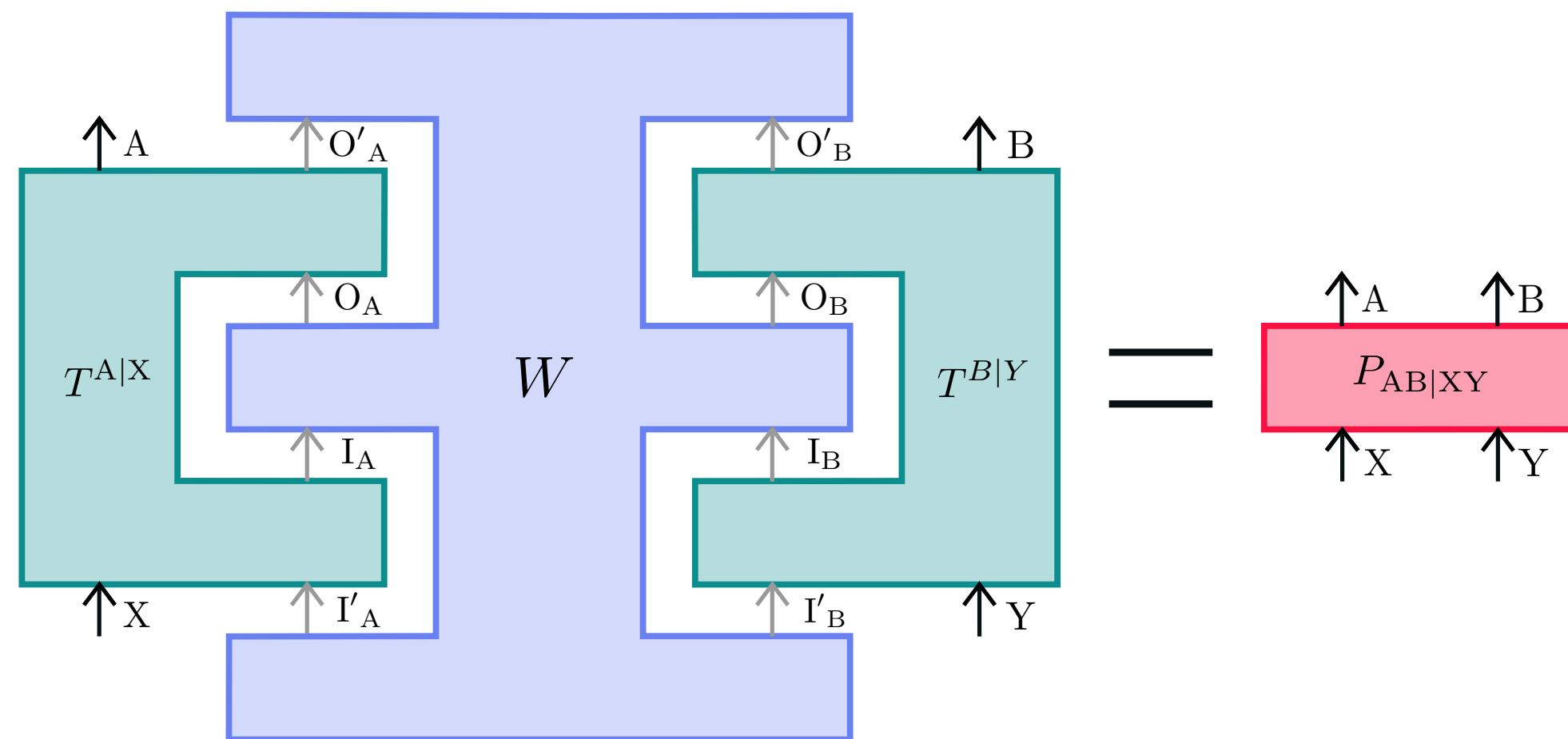


$$W * T^{A|X} * T^{B|Y} = P_{AB|XY}$$

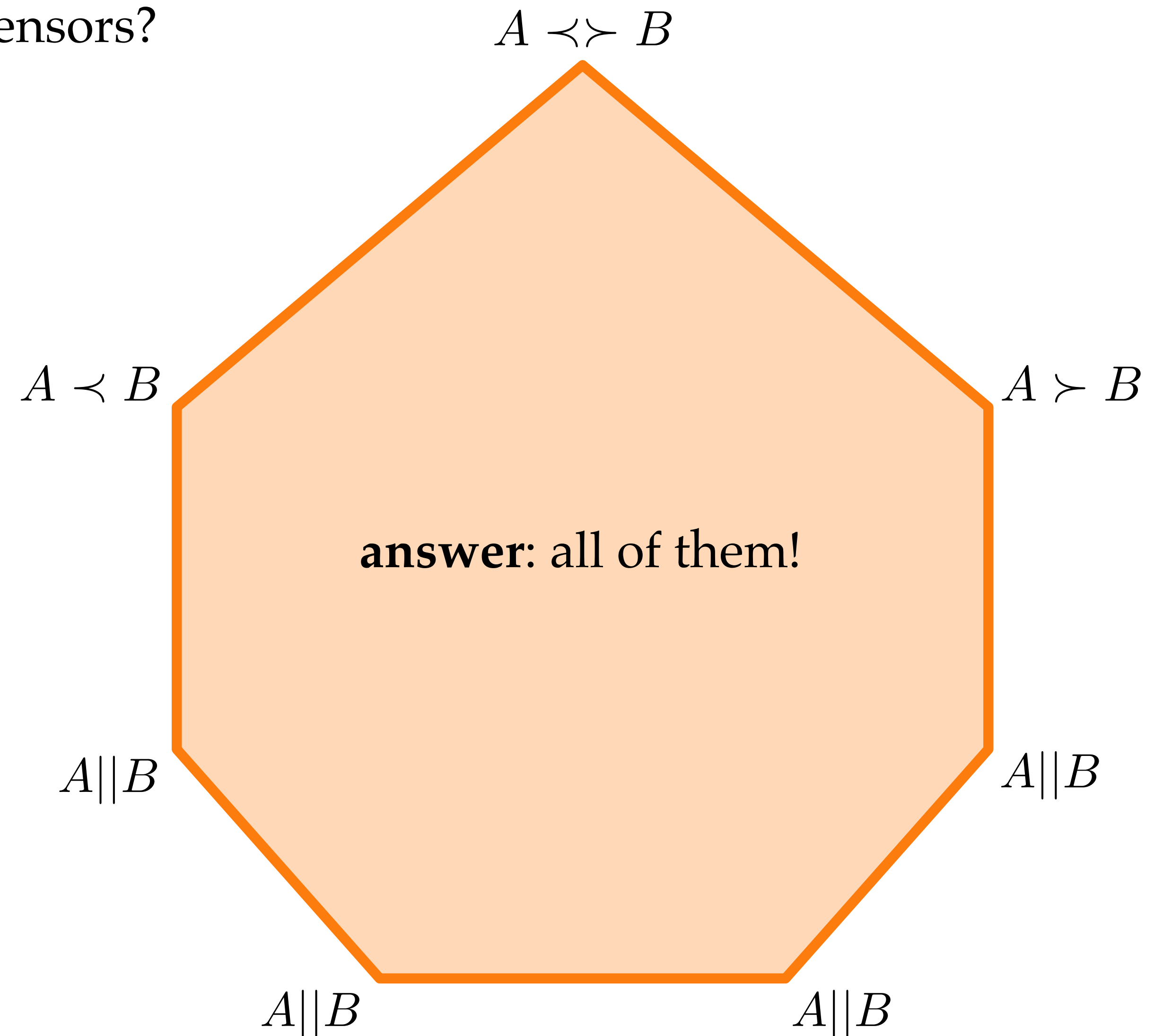


Correlations from process tensors

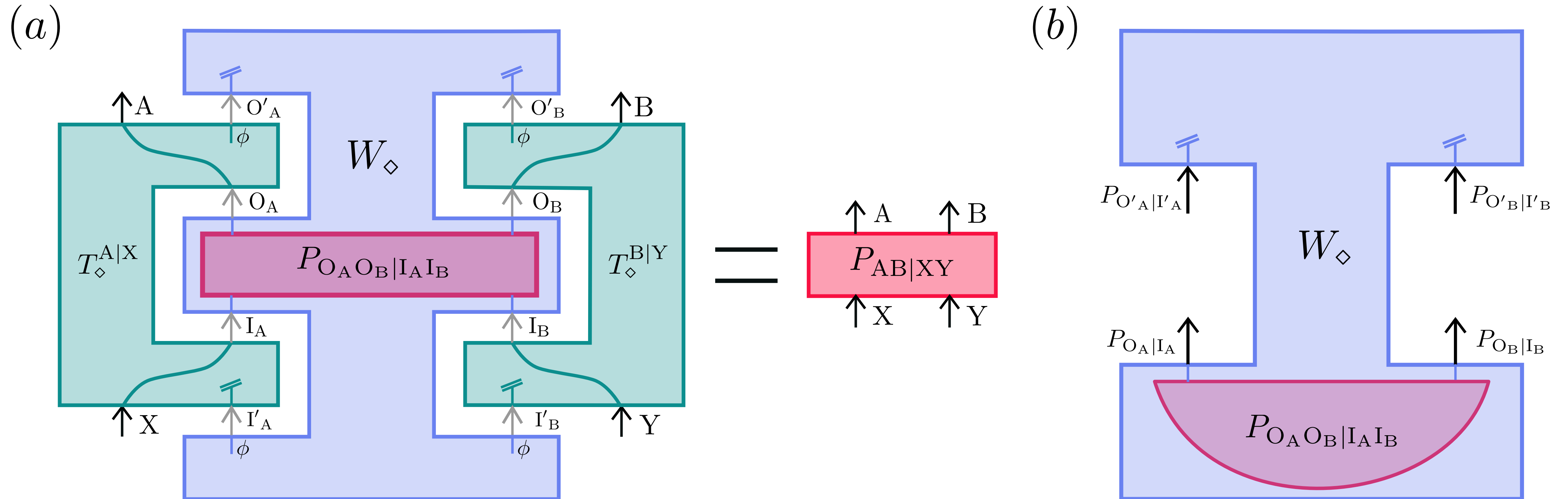
What are the correlations that can be produced by local probabilistic operations acting on process tensors?



$$W * T^{A|X} * T^{B|Y} = P_{AB|XY}$$



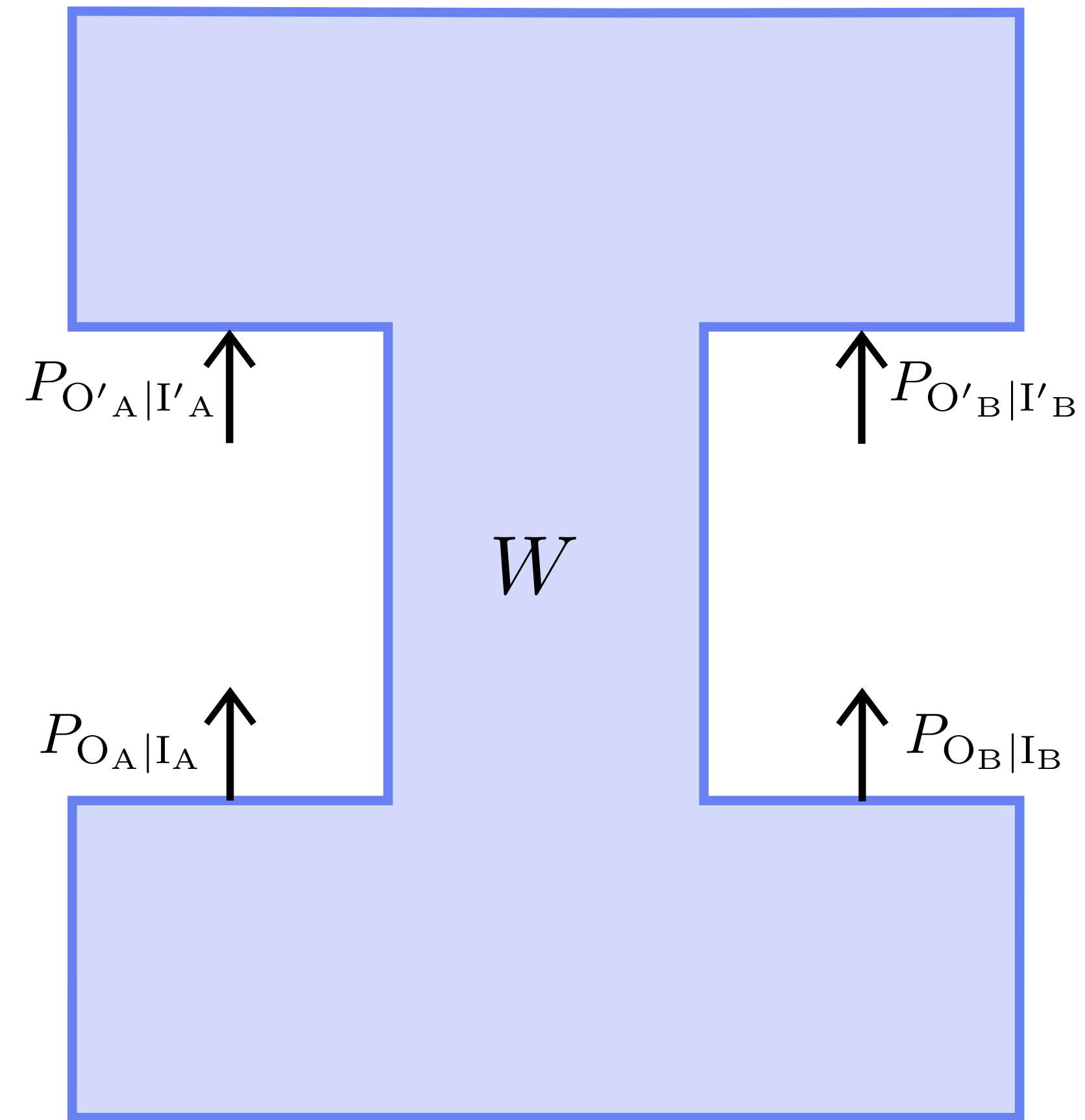
Correlations from process tensors



All correlations can be created as a **state preparation** by the process tensor and simply read by the local operations.

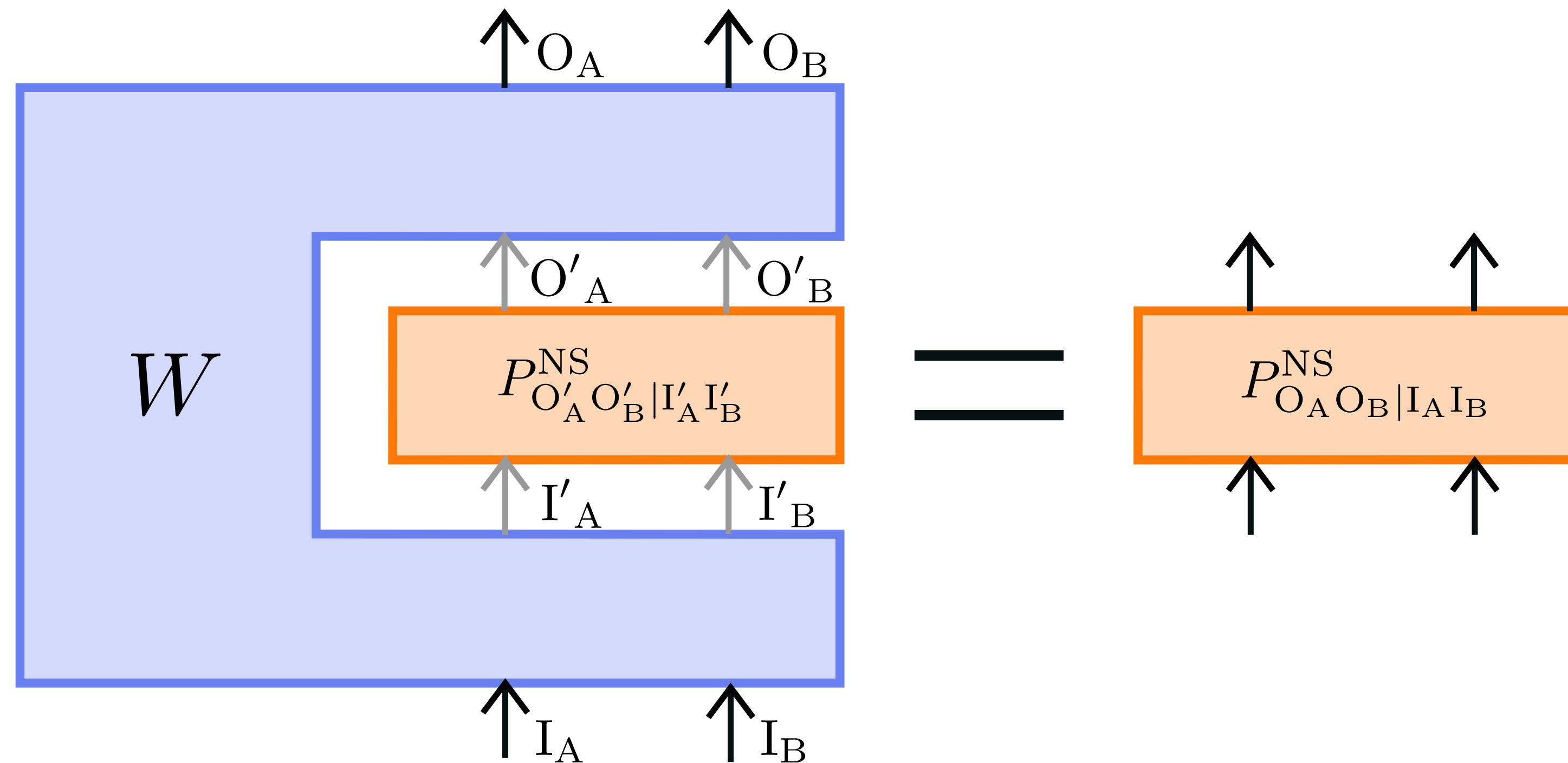
Nonsignalling preservation (NSP)

The principle of nonsignaling preservation



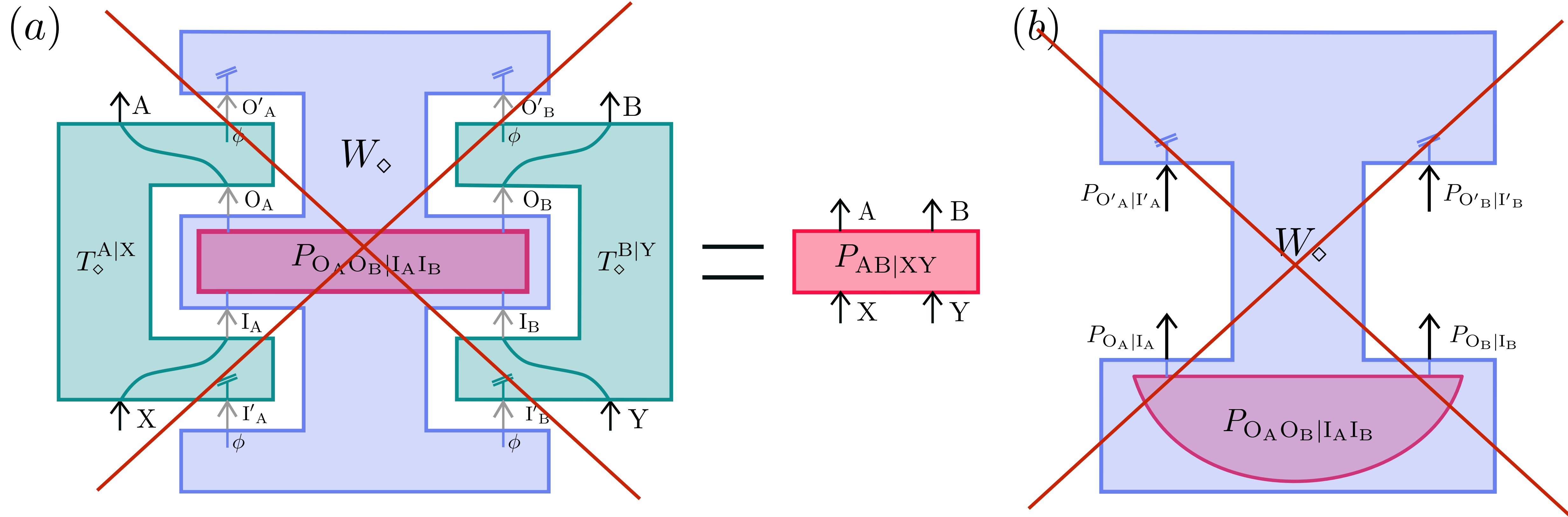
A process tensor (and a process matrix) can be viewed as a channel mapping systems from the output spaces to the input spaces

The principle of nonsignaling preservation



A process tensor that respects NSP must map all nonsignalling boxes in output spaces to nonsignalling boxes in the input spaces

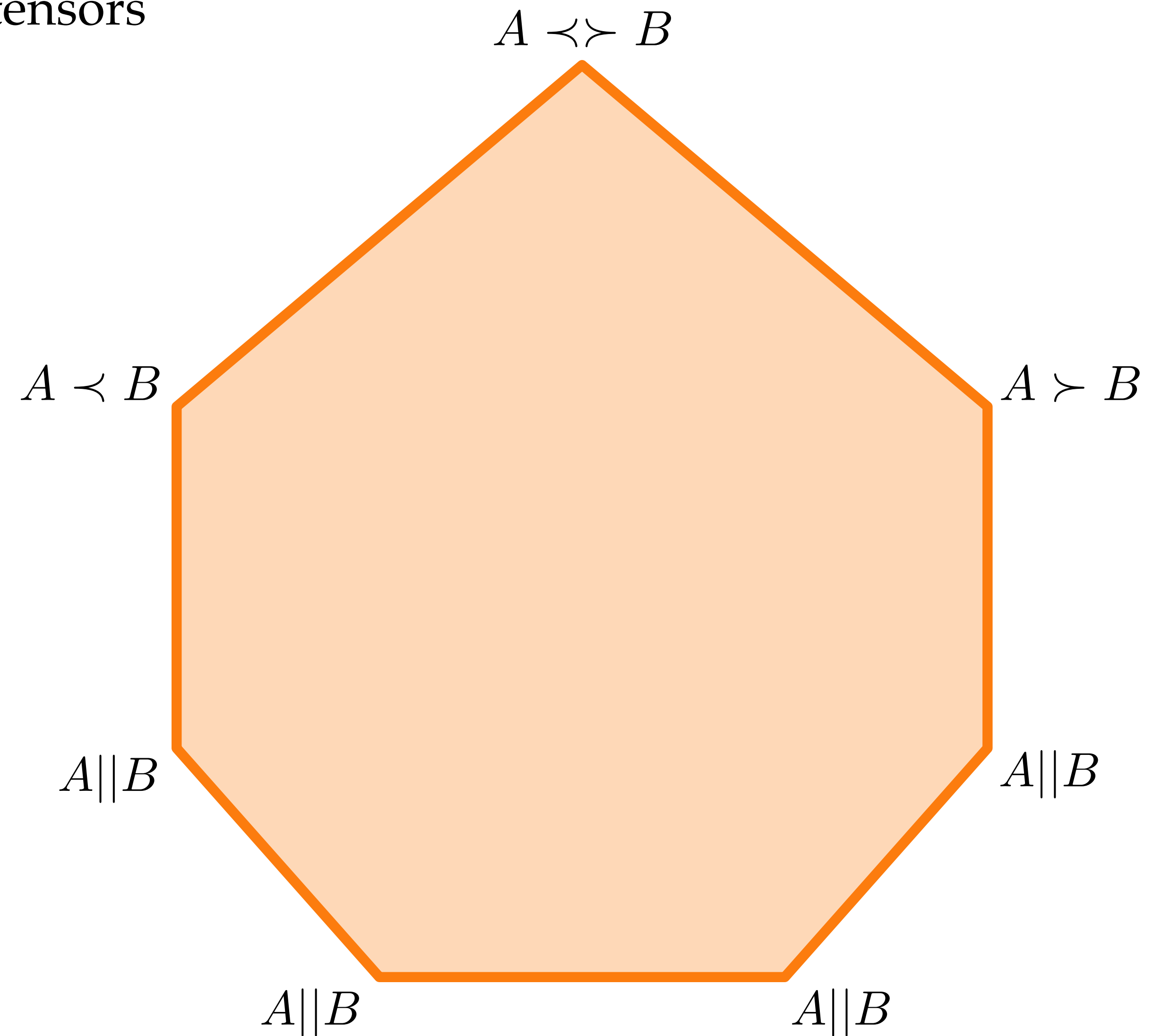
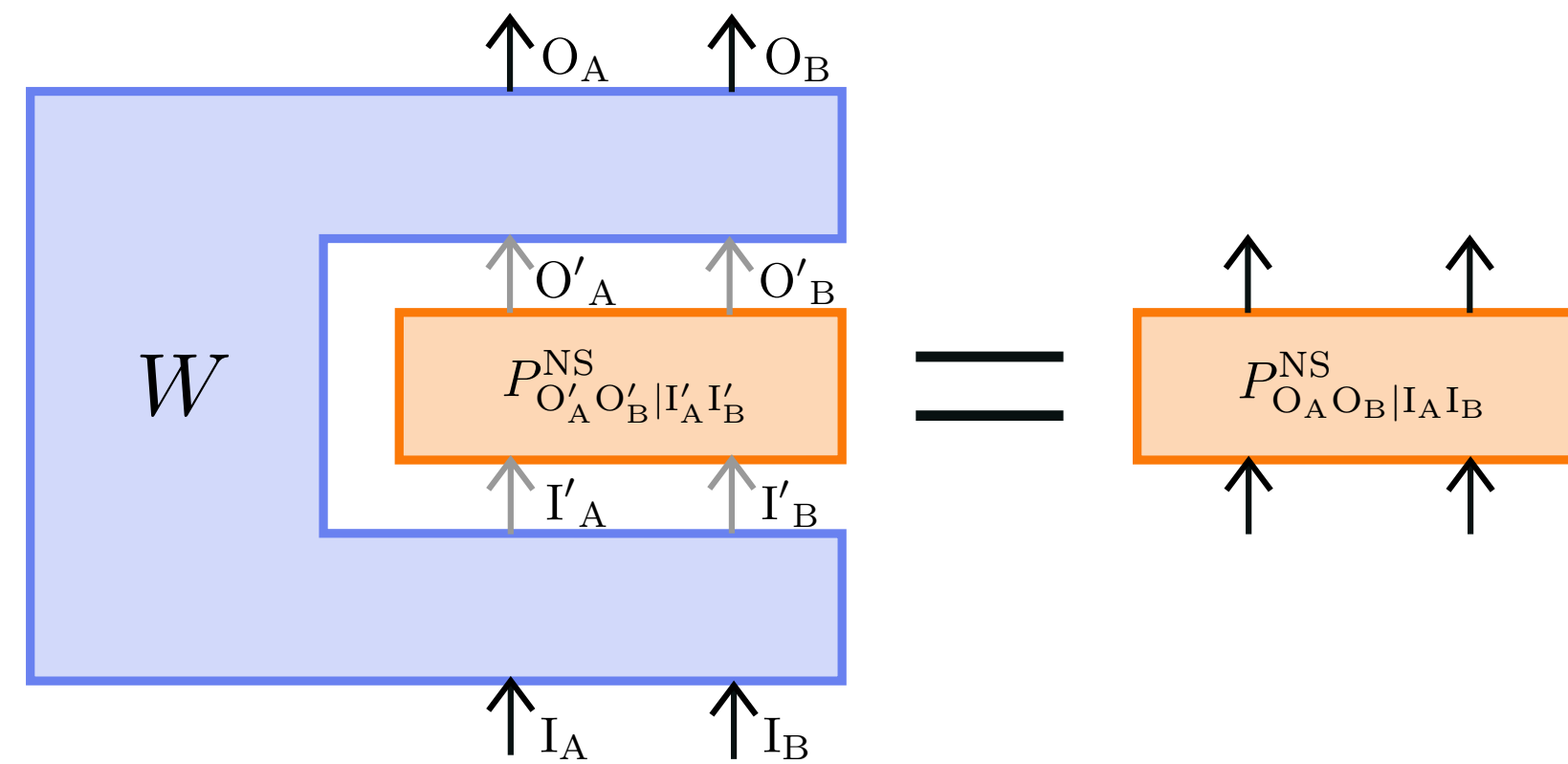
The principle of nonsignaling preservation



This process tensor does not respect NSP

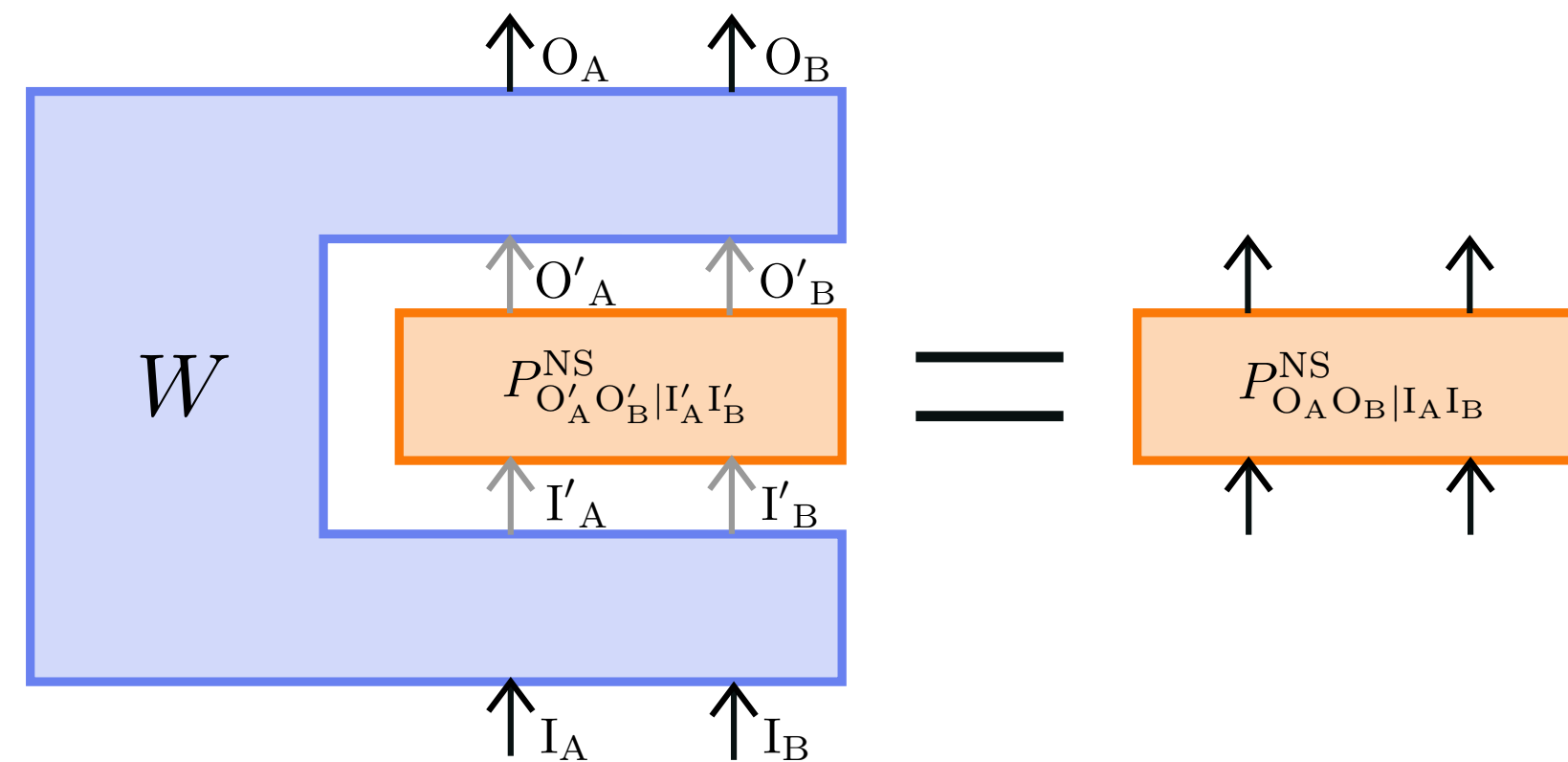
Correlations from process tensors + NSP

What are the correlations that can be produced by local probabilistic operations acting on process tensors that respect NSP?

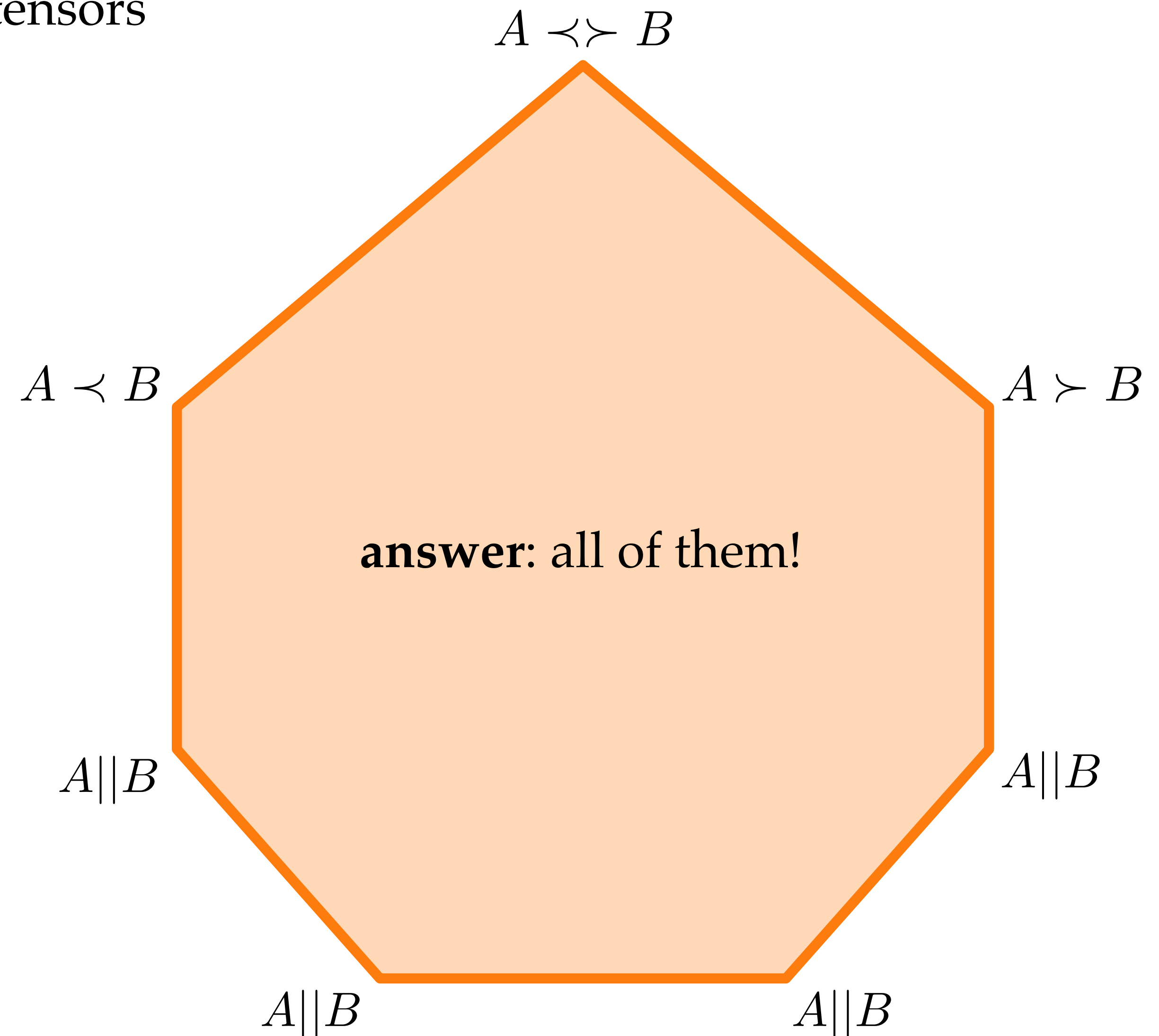


Correlations from process tensors + NSP

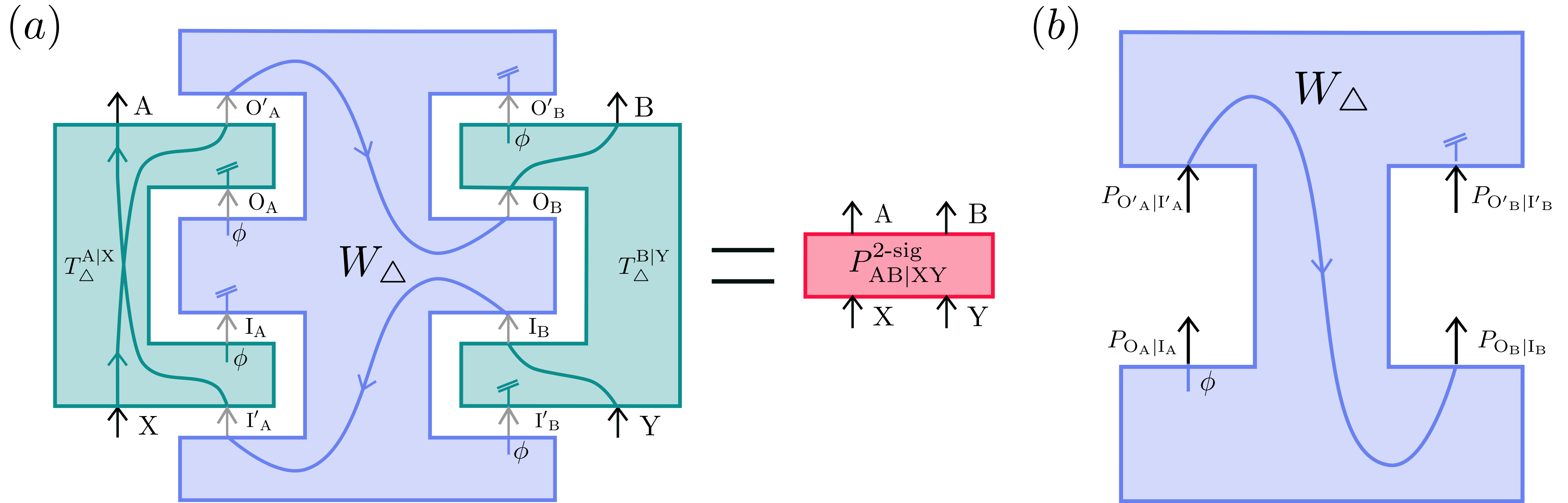
What are the correlations that can be produced by local probabilistic operations acting on process tensors that respect NSP?



Constraints the set of process tensors
but not of the correlations

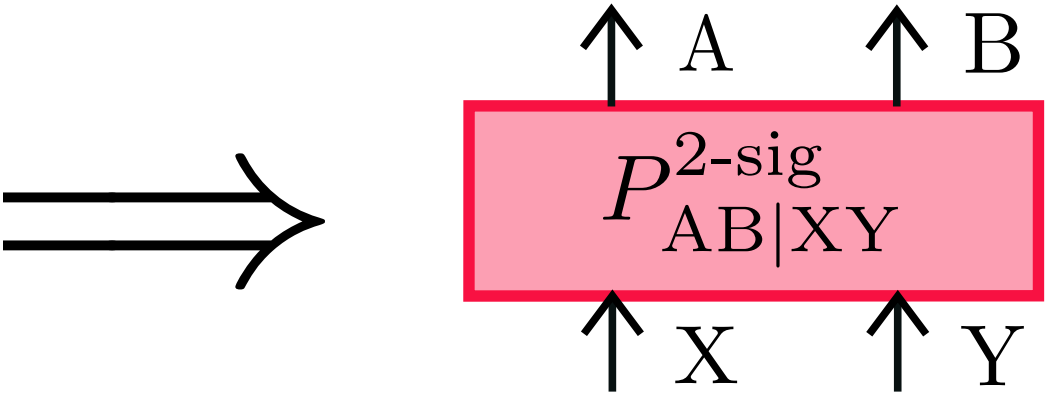
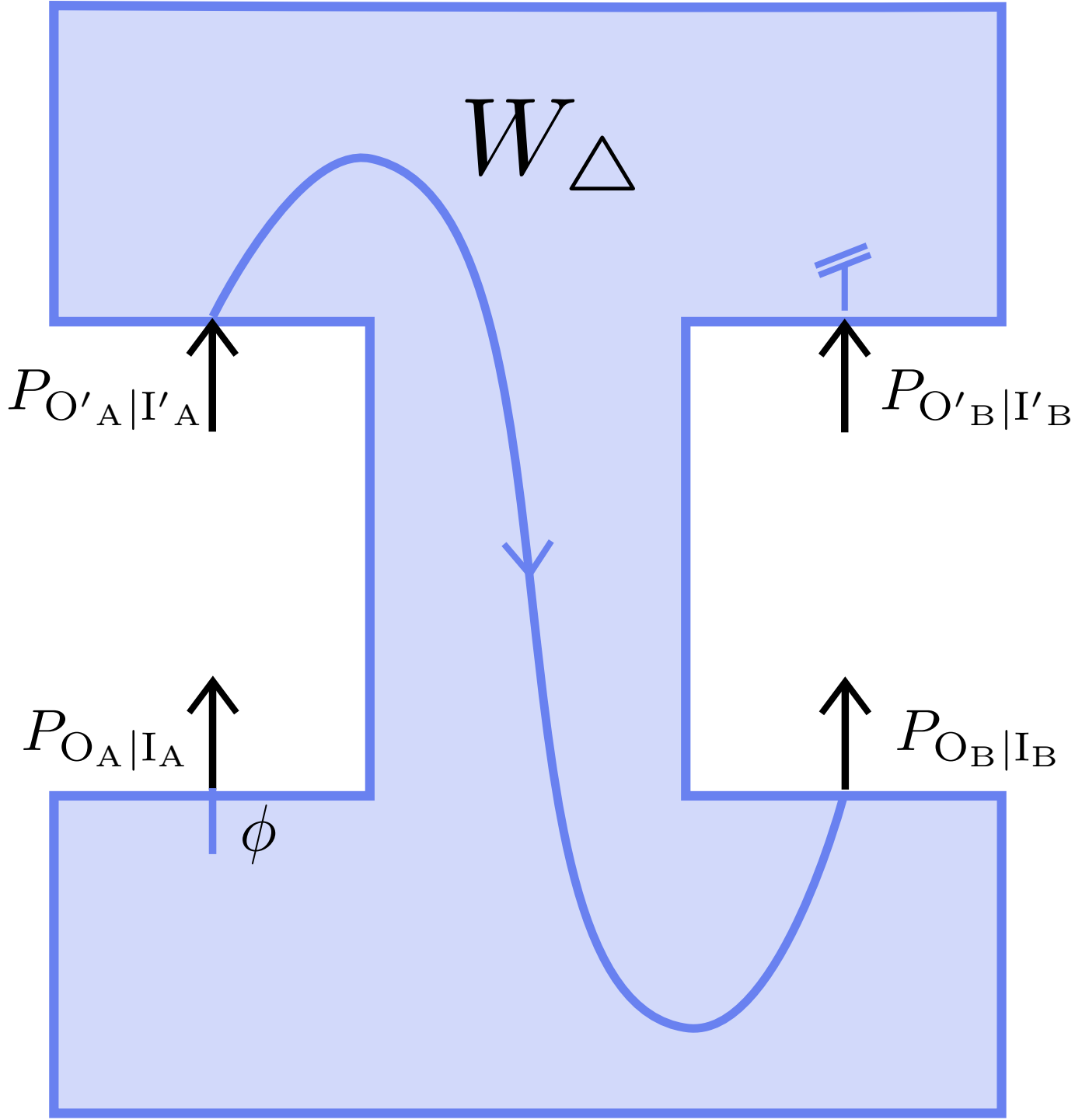
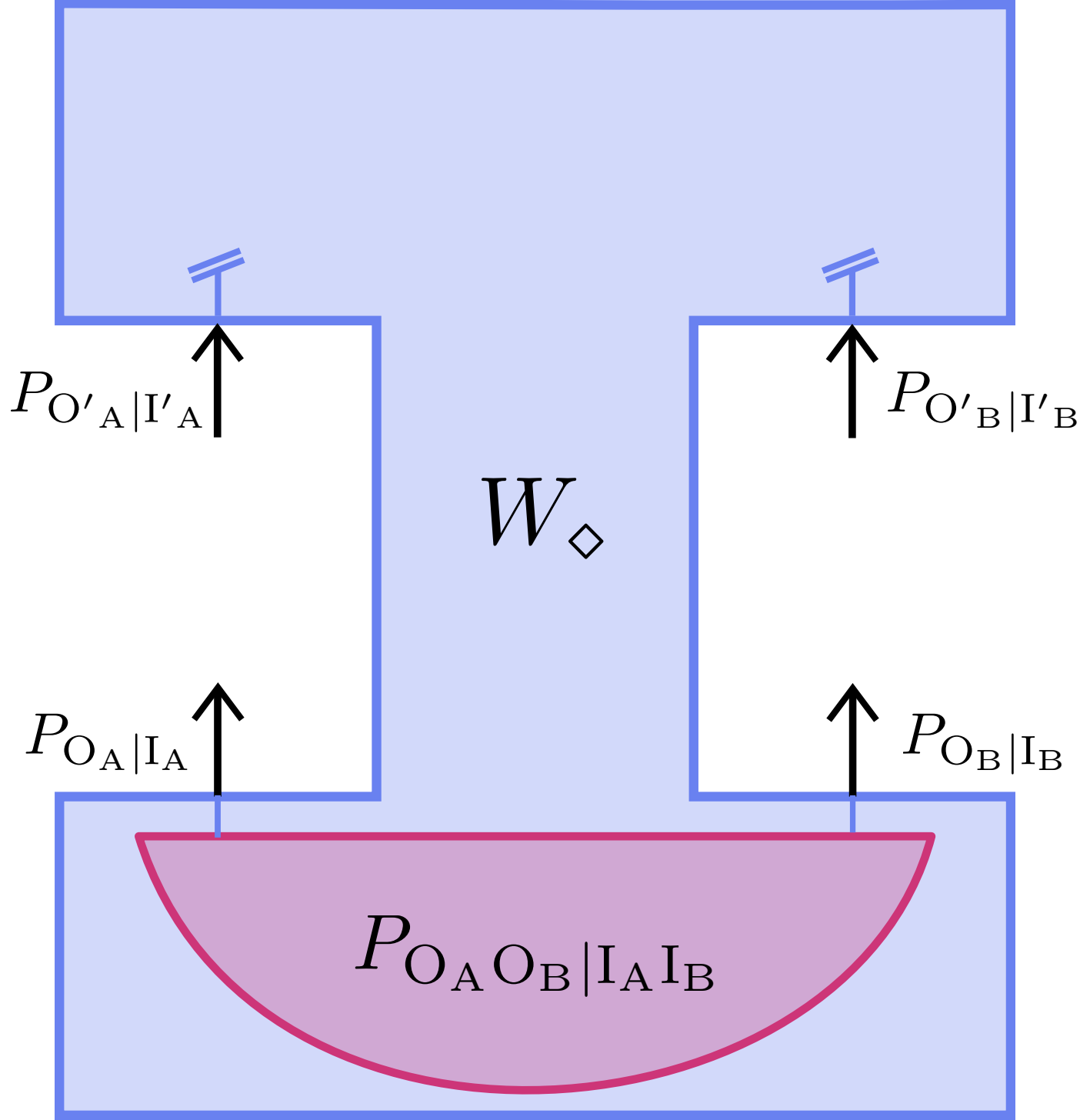


Correlations from process tensors + NSP

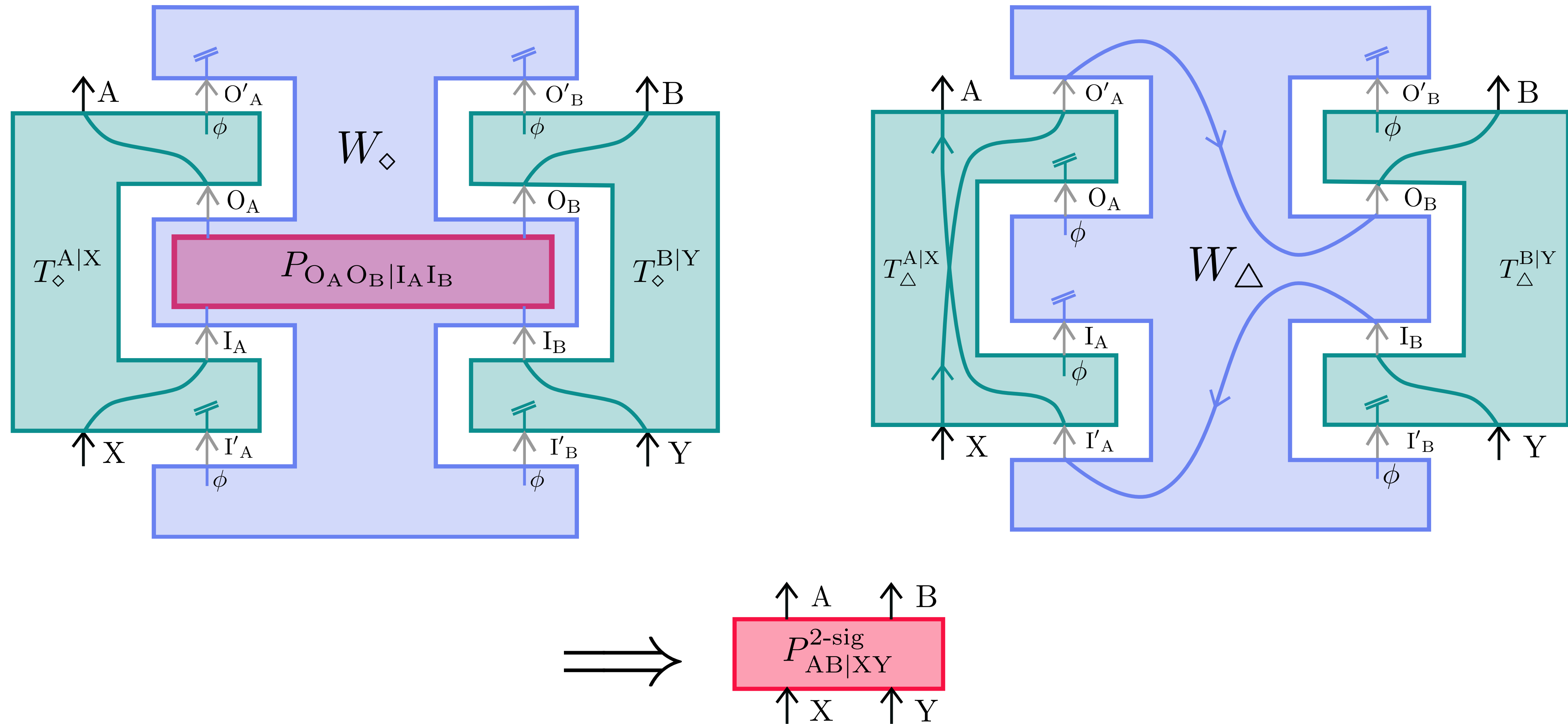


Two-way signalling correlations can be created from a process tensor that respects NSP
and constitutes a **one-way identity channel**

Process tensors that produce two-way signalling



Local operations that produce two-way signalling



No signalling without system exchange (NSWSE)

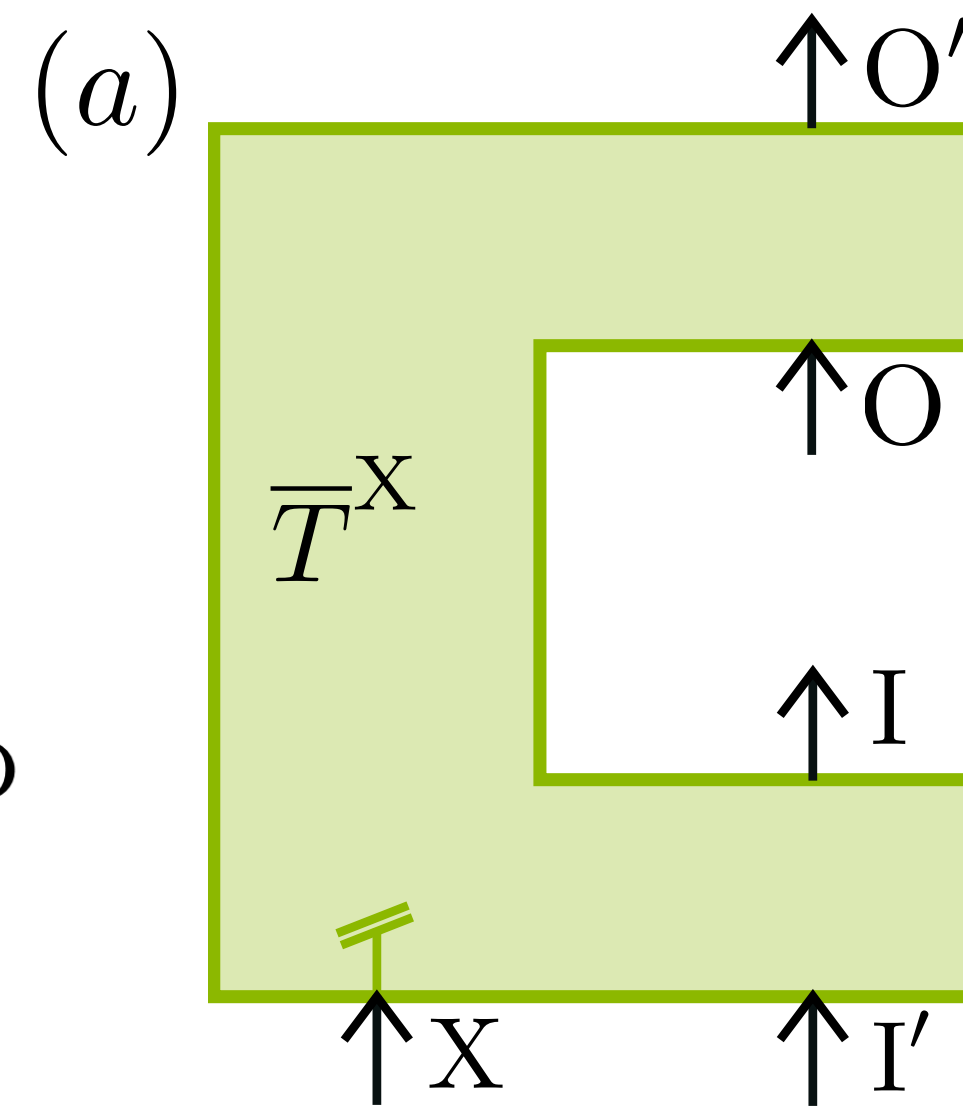
Nonsignalling operations

$$\overline{T}^X * P_{O|I} = P_{O'|I'} \quad \forall \quad X$$

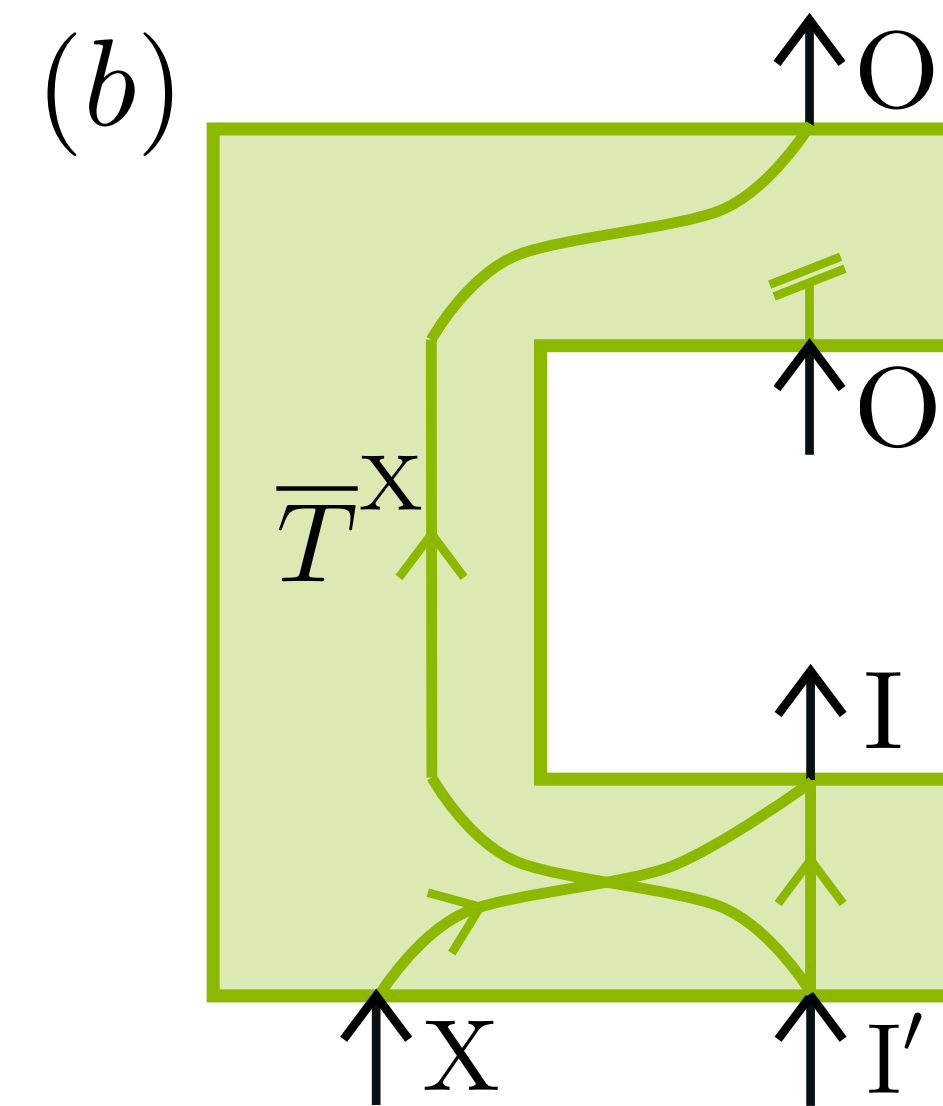
Nonsignalling operations

$$\overline{T}^X * P_{O|I} = P_{O'|I'} \quad \forall X$$

$$\begin{aligned} \overline{T}^X &= P_{I|I'} P_{O'|I'O} \\ &= \sum_{\lambda} \pi_{\lambda} D_{I|I'}^{\lambda} D_{O'|I'O}^{\lambda} \end{aligned}$$



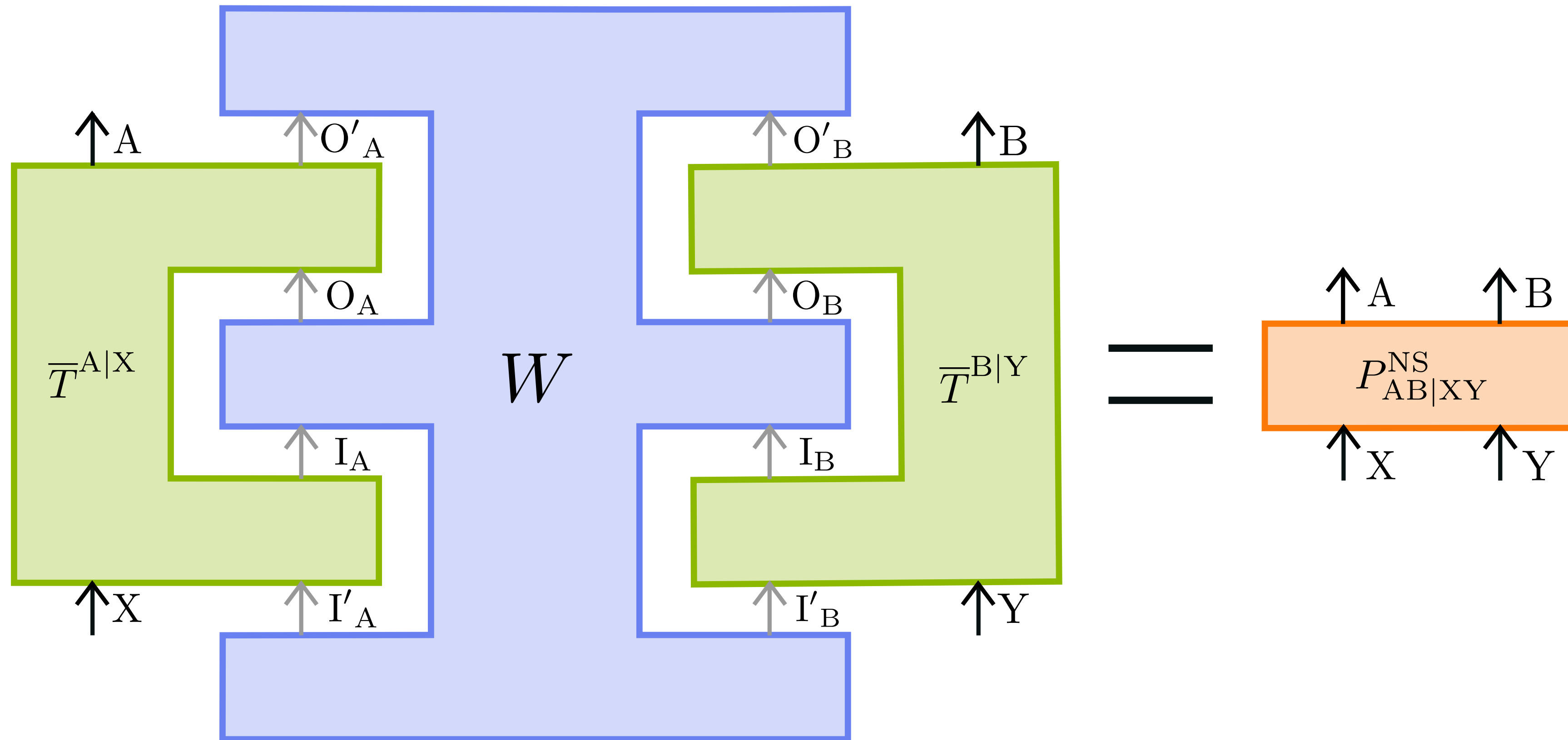
Case 1 (trivial)



Case 2 (nontrivial)

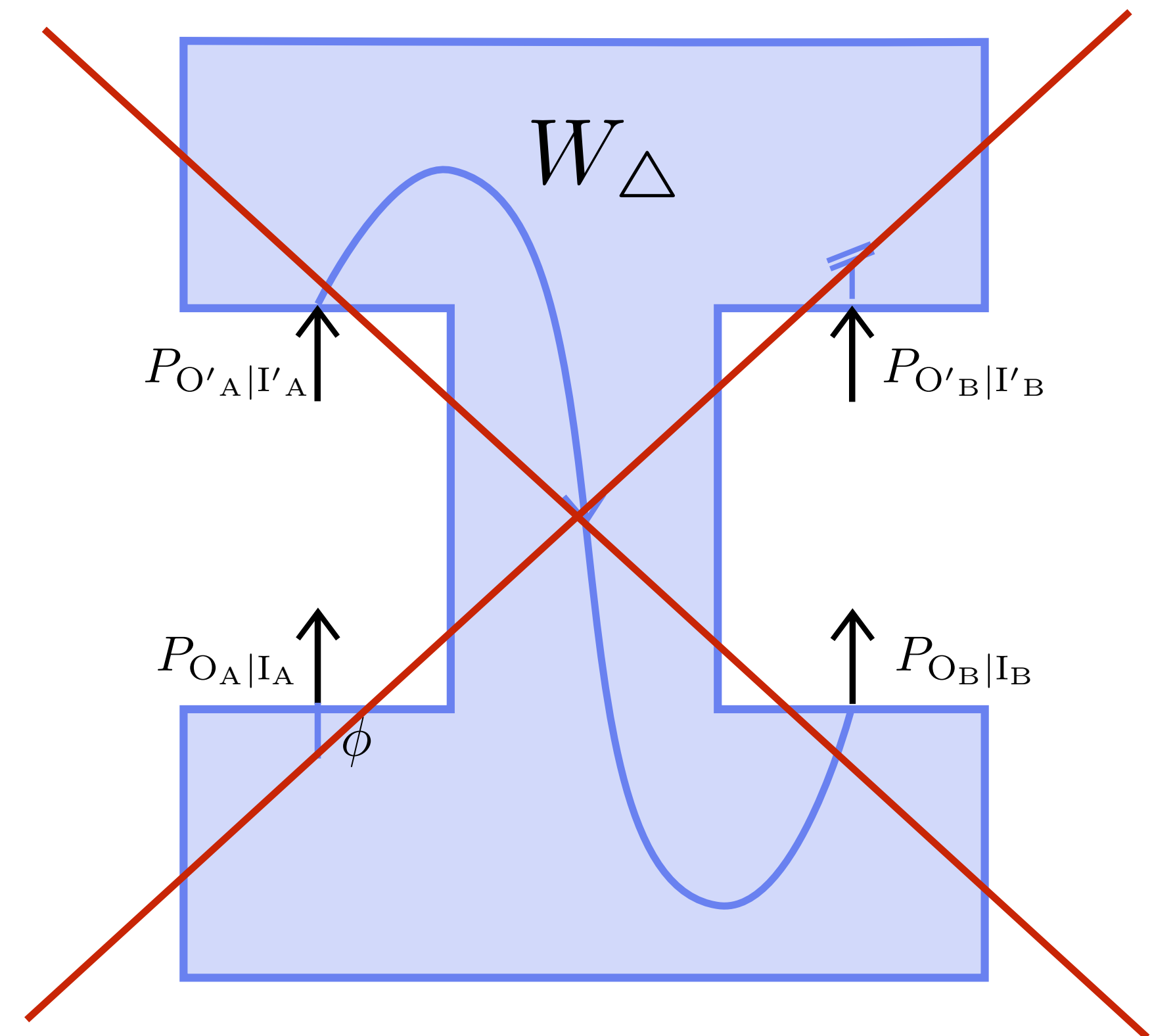
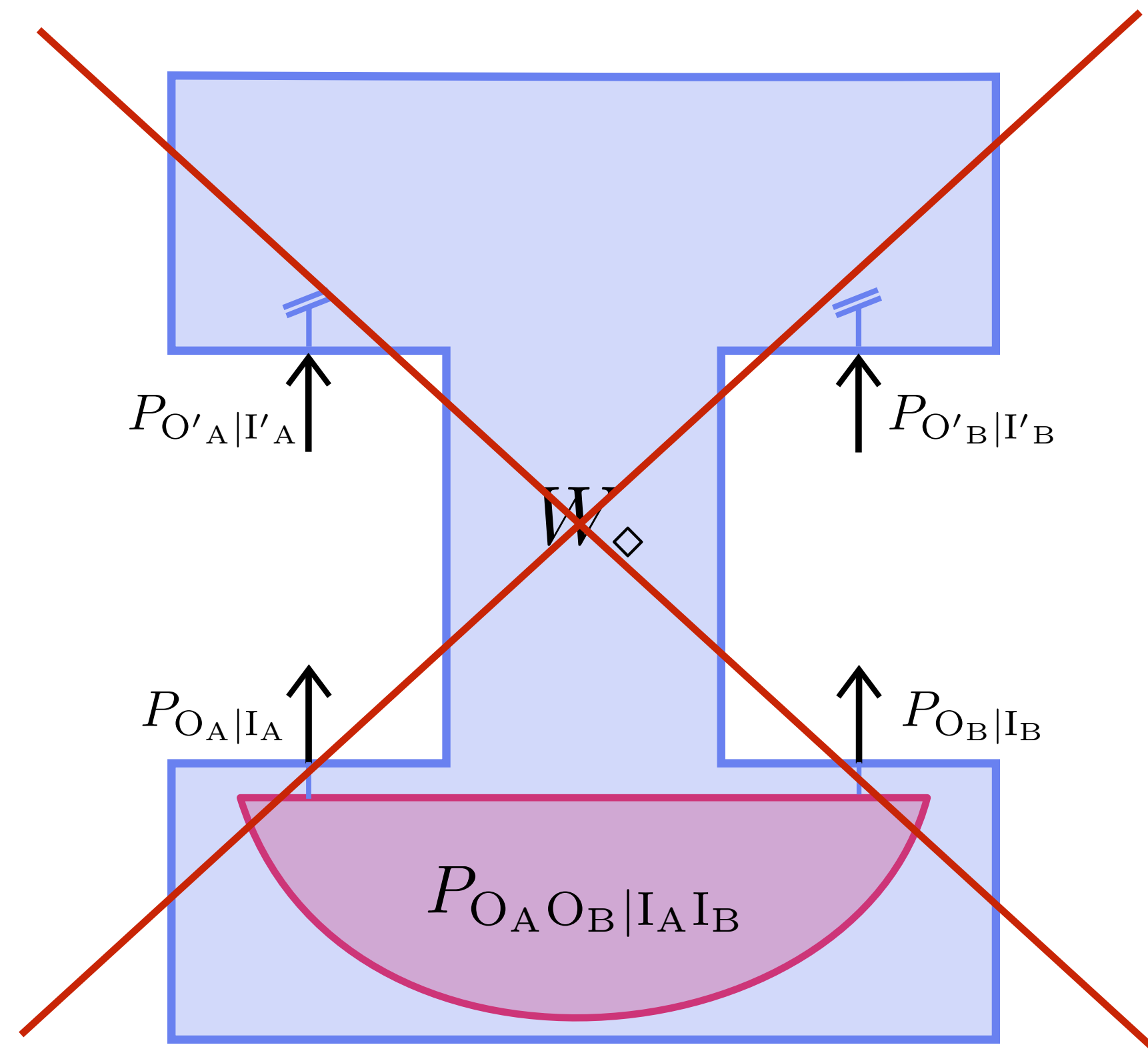
$$\begin{aligned} \overline{T}^X &= P_{I|I'X} P_{O'|I'} \\ &= \sum_{\lambda} \pi_{\lambda} D_{I|I'X}^{\lambda} D_{O'|I'}^{\lambda} \end{aligned}$$

The principle of no signalling without system exchange (NSWSE)



A process tensor that respects NSWSE can create only nonsignalling correlations when acted upon by nonsignalling operations

The principle of no signalling without system exchange (NSWSE)

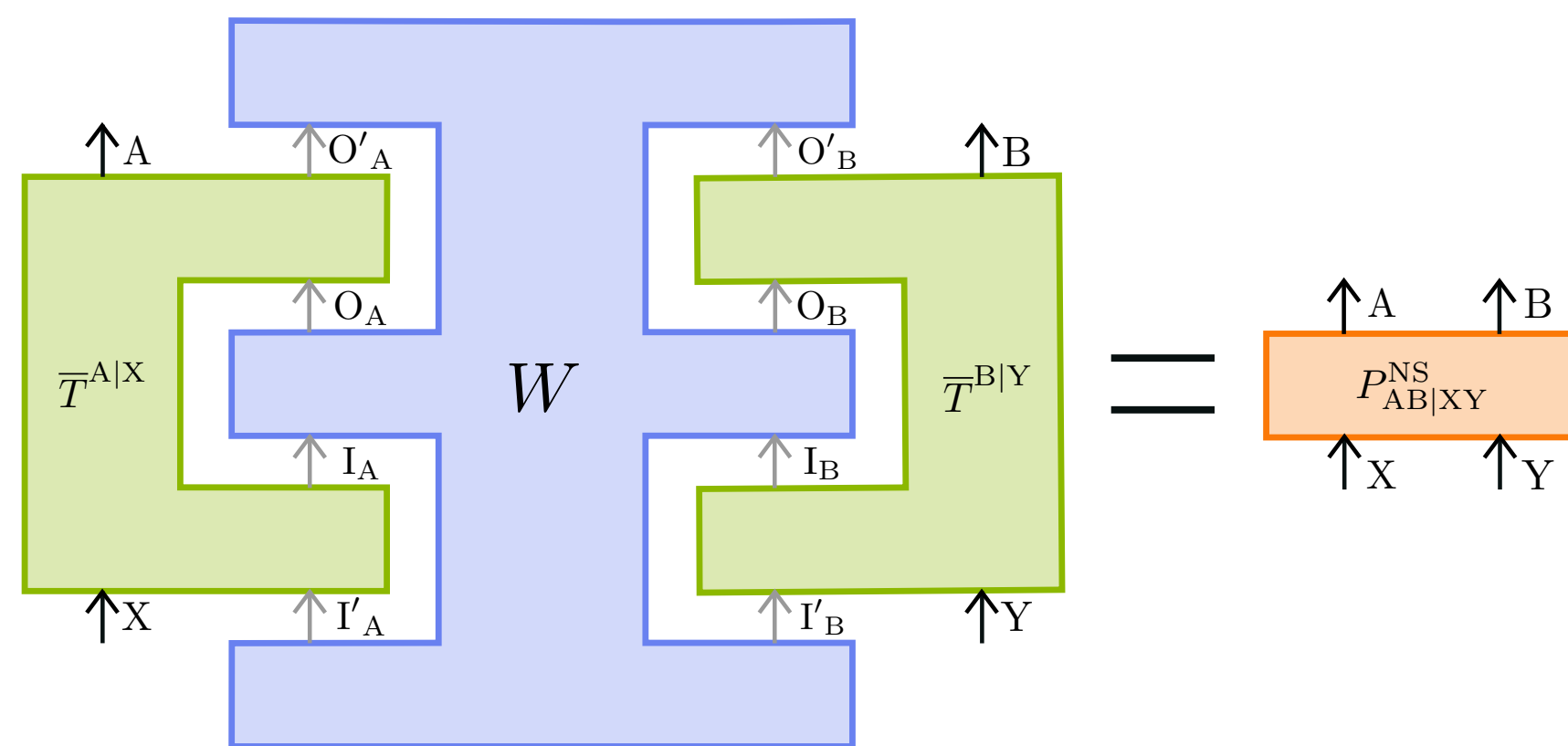


This process tensor does not respect NSWSE

NSWSE is strictly stronger than NSP

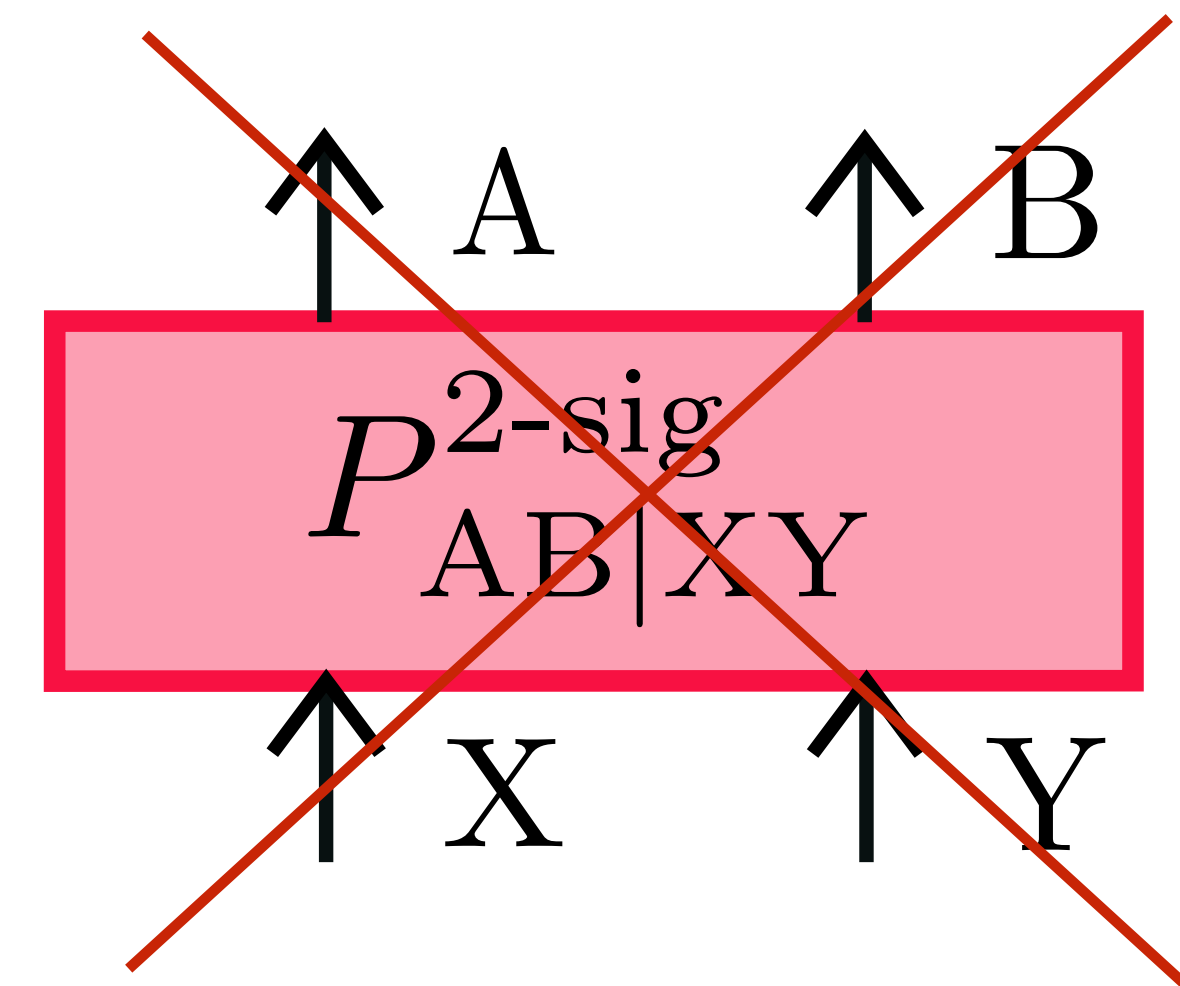
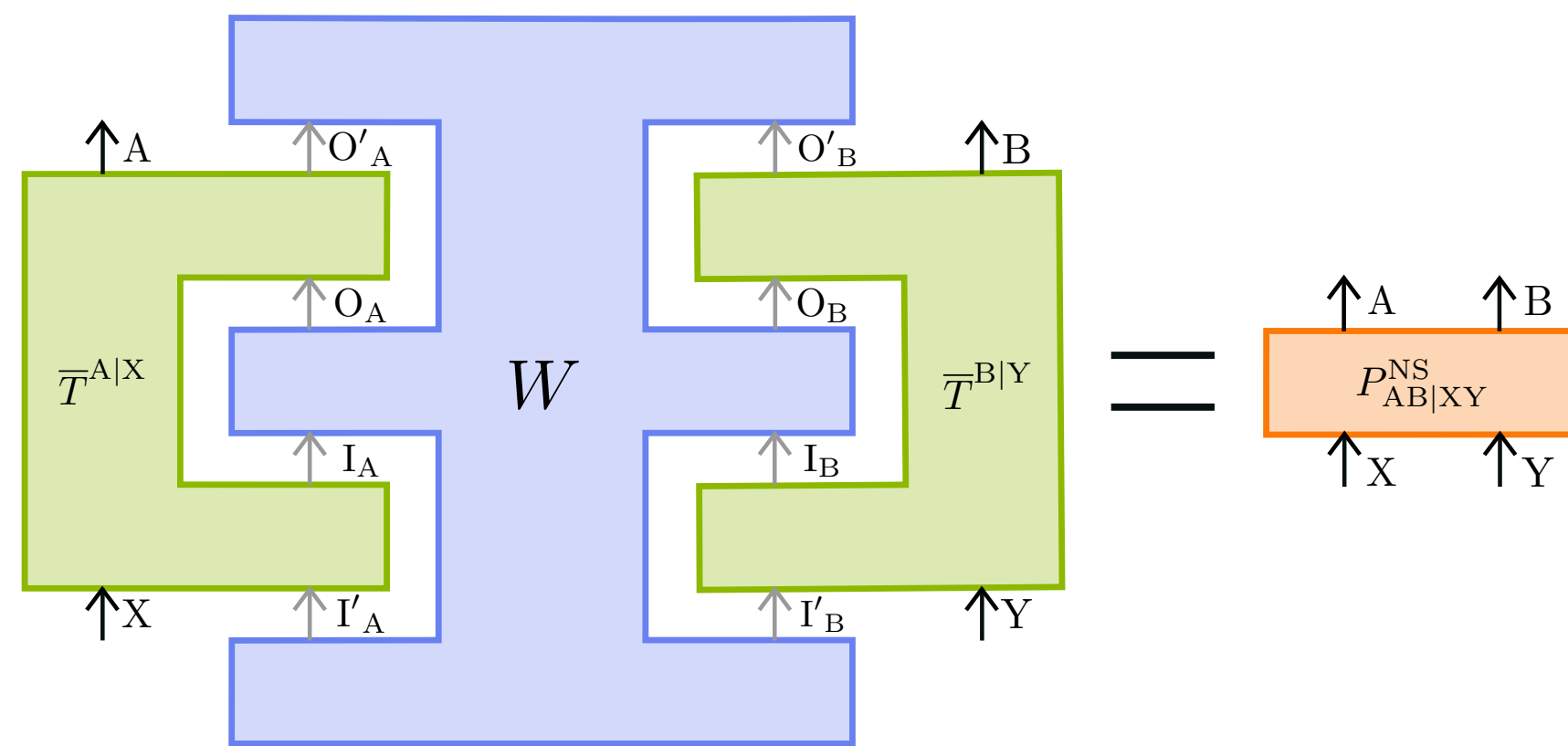
Correlations from process tensors + NSWSE

What are the correlations that can be produced by local probabilistic operations acting on process tensors that respect NSWSE?



Correlations from process tensors + NSWSE

What are the correlations that can be produced by local probabilistic operations acting on process tensors that respect NSWSE?



Constraints the set of process tensors
and the set of correlations

answer: no two-way signalling!

the resulting set of correlations is a polytope

A theory of boxworld processes

Process tensors that respect NSP and NSWSE are called **boxworld processes**

Correlations that can be produced by boxworld processes and local operations
are called **boxworld correlations**

A theory of boxworld processes

- All causal correlations are achieved in boxworld
- There exist noncausal boxworld correlations (genuine two-way signalling)
- Boxworld correlations form a polytope (in finite dimensions)
- There is no perfect two-way signalling in boxworld correlations (in finite dimensions)

Comparison to process matrix correlations

Causal inequalities

$$\text{GYNI}(P) := \frac{1}{4} \sum_{A,B,X,Y} \delta_{A,Y} \delta_{B,X} P_{AB|XY} \leq \frac{1}{2}$$

$$\text{LGYNI}(P) := \frac{1}{4} \sum_{A,B,X,Y} \delta_{X(A \oplus Y),0} \delta_{Y(B \oplus X),0} P_{AB|XY} \leq \frac{3}{4}$$

$$\text{OCB}(P) := \frac{1}{8} \sum_{\substack{A,B, \\ X,Y,Y'}} \delta_{(Y' \oplus 1)(A \oplus Y),0} \delta_{Y'(B \oplus X),0} P_{AB|XY Y'} \leq \frac{3}{4}$$

Almeida et al., PRL 104, 230404 (2010)

Branciard et al., NJP 18, 013008 (2016)

Oreshkov et al., Nat. Commun. 3, 1092 (2012)

Causal inequalities

Expression	Causal
GYNI	$\leq \frac{1}{2} = 0.5$
LGYNI	$\leq \frac{3}{4} = 0.75$
OCB	$\leq \frac{3}{4} = 0.75$

Causal inequalities

Expression	Causal	Process matrix
GYNI	$\leq \frac{1}{2} = 0.5$	≤ 0.7592
LGYNI	$\leq \frac{3}{4} = 0.75$	≈ 0.8194
OCB	$\leq \frac{3}{4} = 0.75$	$= \frac{2+\sqrt{2}}{4} \approx 0.8536$

Liu and Chiribella
arXiv:2403.02749 (2024)

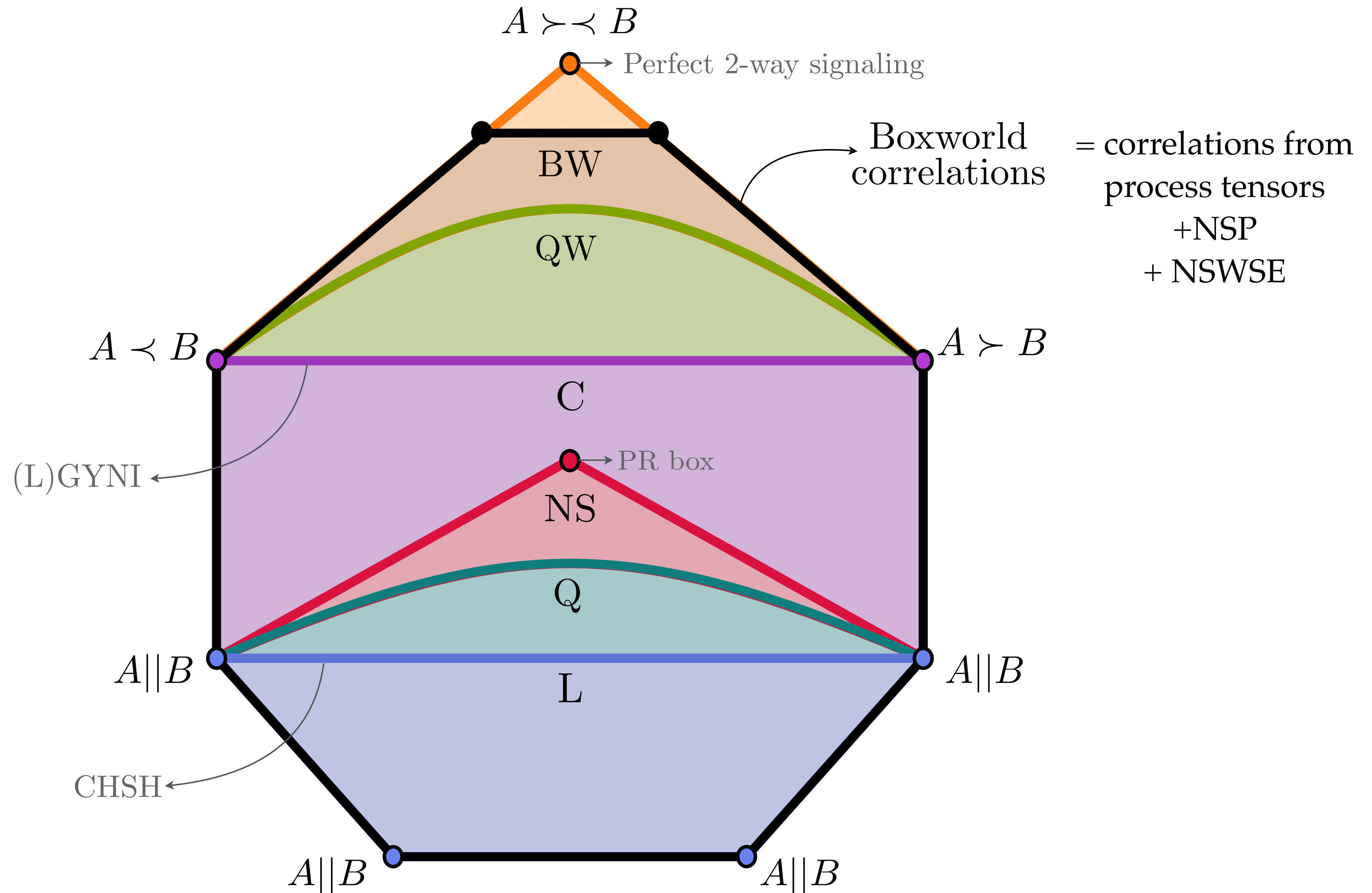
Causal inequalities

Expression	Causal	Process matrix	Boxworld
GYNI	$\leq \frac{1}{2} = 0.5$	≤ 0.7592	$\geq \frac{3}{4} = 0.75$
LGYNI	$\leq \frac{3}{4} = 0.75$	≈ 0.8194	$\geq \frac{11}{12} \approx 0.9167$
OCB	$\leq \frac{3}{4} = 0.75$	$= \frac{2+\sqrt{2}}{4} \approx 0.8536$	$= 1$

Causal inequalities

Expression	Causal	Process matrix	Boxworld	local dim
GYNI	$\leq \frac{1}{2} = 0.5$	≤ 0.7592	$\geq \frac{3}{4} = 0.75$	$d = 3$
LGYNI	$\leq \frac{3}{4} = 0.75$	≈ 0.8194	$\geq \frac{11}{12} \approx 0.9167$	$d = 2$
OCB	$\leq \frac{3}{4} = 0.75$	$= \frac{2+\sqrt{2}}{4} \approx 0.8536$	$= 1$	$d = 2$

Conjecture



Conclusions

- Main contributions

- Fully fledged higher-order boxworld theory with indefinite causal order
- Conjecture of an outer approximation of the set of process matrix correlations

- Open questions

- Determine the validity of this conjecture
- Find information-theoretic and physical principles that can be imposed at the level of the correlations

Thank you!

Extra

Process tensor

$$W \geq 0$$

$$r(W) = d_{I_A} d_{O'_A} d_{I_B} d_{O'_B}$$

$$\mathbb{A}(1 - O'_B + O_B O'_B - I_B O_B O'_B) W = 0$$

$$(1 - O'_A + O_A O'_A - I_A O_A O'_A) \mathbb{B} W = 0$$

$$(1 - O'_A + O_A O'_A - I_A O_A O'_A)(1 - O'_B + O_B O'_B - I_B O_B O'_B) W = 0,$$

$$\begin{aligned}
W_{\diamond} &= \delta_{I'_{\text{A}},\phi} \delta_{I'_{\text{B}},\phi} P_{\text{O}_{\text{A}}\text{O}_{\text{B}}|\text{I}_{\text{A}}\text{I}_{\text{B}}} \\
T_{\diamond}^{\text{A}|\text{X}} &= \delta_{\text{I}_{\text{A}},\text{X}} \delta_{\text{O}'_{\text{A}},\phi} \delta_{\text{A},\text{O}_{\text{A}}} \\
T_{\diamond}^{\text{B}|\text{Y}} &= \delta_{\text{I}_{\text{B}},\text{Y}} \delta_{\text{O}'_{\text{B}},\phi} \delta_{\text{B},\text{O}_{\text{B}}},
\end{aligned}$$

Nonsignalling preservation

$$O_A(O'_A O'_B I'_A I'_B)W = O_A I_A(O'_A O'_B I'_A I'_B)W$$

$$O_B(O'_A O'_B I'_A I'_B)W = O_B I_B(O'_A O'_B I'_A I'_B)W$$

$$\begin{aligned} O_A(1 - O'_A + O'_A I'_A - O'_B O'_A I'_A)W \\ = O_A I_A(1 - O'_A + O'_A I'_A - O'_B O'_A I'_A)W \end{aligned}$$

$$\begin{aligned} O_B(1 - O'_B + O'_B I'_B - O'_A O'_B I'_B)W \\ = O_B I_B(1 - O'_B + O'_B I'_B - O'_A O'_B I'_B)W. \end{aligned}$$

$$\begin{aligned}
W_{\Delta} &= \delta_{I'_{\text{A}}, I_{\text{B}}} \delta_{O_{\text{A}}, \phi} \delta_{I'_{\text{B}}, \phi} \delta_{O_{\text{B}}, O'_{\text{A}}} \\
T_{\Delta}^{\text{A}|\text{X}} &= \delta_{I_{\text{A}}, \phi} \delta_{O'_{\text{A}}, \text{X}} \delta_{\text{A}, I'_{\text{A}}}, \\
T_{\Delta}^{\text{B}|\text{Y}} &= \delta_{I_{\text{B}}, \text{Y}} \delta_{O'_{\text{B}}, \phi} \delta_{\text{B}, O_{\text{B}}}.
\end{aligned}$$

No signalling without system exchange

$$O_A W = O_A I_A W$$

$$O_B W = O_B I_B W,$$

Boxworld process

$$W \geq 0$$

$$r(W) = d_{I_A} d_{O'_A} d_{I_B} d_{O'_B}$$

$${}_A W = {}_A O'_B W$$

$${}_B W = {}_{O'_A} B W$$

$$W = {}_{O'_A} W + {}_{O'_B} W - {}_{O'_A O'_B} W$$

$${}_{O_A} W = {}_{I_A} O_A W$$

$${}_{O_B} W = {}_{I_B} O_B W,$$

GYNI = 2/3

The value of 2/3 for GYNI is obtained by a process that, conditioned on O'_A, O'_B produces a $(\frac{1}{3}, \frac{2}{3})$ mixture of a constant box $P_{O_A O_B | I_A I_B} = P_{O_A O_B}$ and a PR box [40, 41, 45], given by

$$W_{\text{GYNI}} = \frac{1}{3} \delta_{O_A, O'_B} \delta_{O_B, O'_A} + \frac{2}{3} \delta_{(O_A \oplus O_B \oplus O'_A \oplus O'_B \oplus 1), ((I_A \oplus O'_A \oplus 1)(I_B \oplus O'_B \oplus 1))} / 2, \quad (\text{B45})$$

$$(\text{B46})$$

and the following local operations:

$$T_{\text{GYNI}}^{A|X} = \delta_{I_A, X} \delta_{O'_A, X} \delta_{A, O_A} \quad (\text{B47})$$

$$T_{\text{GYNI}}^{B|Y} = \delta_{I_B, Y} \delta_{O'_B, Y} \delta_{B, O_B}. \quad (\text{B48})$$

Combined, they lead to the correlations

$$\overline{P}_{AB|XY}^{(\text{GYNI}=2/3)} = \frac{1}{3} \delta_{A, Y} \delta_{B, X} + \frac{2}{3} \left(\frac{\delta_{A, Y} \delta_{B, X}}{2} + \frac{\delta_{A \oplus 1, Y} \delta_{B \oplus 1, X}}{2} \right). \quad (\text{B49})$$

GYNI = 3/4

1/4	1/2	0	1/2
1/4	0	1/2	0
0	0	0	0
1/4	1/2	1/2	1/2
1/4	0	0	0
0	0	0	0

1/4	1/2	1/2	1
1/4	0	1/2	0
0	0	0	0
1/4	1/2	0	0
1/4	0	0	0
0	0	0	0

1/4	1/2	1/2	1/2
1/4	0	0	0
0	0	0	0
1/4	1/2	0	1/2
1/4	0	1/2	0
0	0	0	0

1/4	0	0	0
1/4	0	1/2	0
0	1/2	0	1/2
1/4	1/2	1/2	1/2
1/4	0	0	0
0	0	0	0

1/4	0	1/2	1/2
1/4	0	1/2	0
0	1/2	0	1/2
1/4	1/2	0	0
1/4	0	0	0
0	0	0	0

1/4	0	1/2	0
1/4	0	0	0
0	1/2	0	1/2
1/4	1/2	0	1/2
1/4	0	1/2	0
0	0	0	0

1/4	1/2	0	0
1/4	0	1/2	0
0	0	0	1/2
1/4	0	1/2	1/2
1/4	0	0	0
0	1/2	0	0

1/4	1/2	1/2	1/2
1/4	0	1/2	0
0	0	0	1/2
1/4	0	0	0
1/4	0	0	0
0	1/2	0	0

1/4	1/2	1/2	0
1/4	0	0	0
0	0	0	1/2
1/4	0	0	1/2
1/4	0	1/2	0
0	1/2	0	0

Table II. Boxworld process for obtaining the value $3/4$ in GYNI. The table in row r and column c is for $O'_A = r, O'_B = c$. Within each table, the columns represent the inputs (I_A, I_B), and the rows the outputs (O_A, O_B).

$$\text{GYNI} = 3/4$$

The corresponding local operations are

XO_A	$AO'_A I_A$		YO_B	$BO'_B I_B$	
00	110	for Alice, and	00	100	for Bob.
01	120		01	120	
10	101		02	000	
11	021		10	111	
			11	111	
			12	001	

That boxworld process in Table II combined with these local operations produce the following correlations:

$$\overline{P}_{AB|XY}^{(\text{GYNI}=3/4)} = \delta_{A,X(Y\oplus 1)\oplus 1} \delta_{B,(X\oplus 1)Y\oplus 1}.$$

LGYNl = 11/12

$$W_{\text{LGYNl}} = \frac{1}{3}P_{\text{LGYNl}}^{\alpha} + \frac{2}{3}P_{\text{LGYNl}}^{\text{PR}} \quad (\text{B52})$$

with

$$P_{\text{LGYNl}}^{\alpha} = \delta_{O_A, I_A} \delta_{O'_B, I_B} \delta_{O'_A} \quad (\text{B53})$$

$$P_{\text{LGYNl}}^{\text{PR}} = \begin{cases} \frac{1}{2}\delta_{O_A \oplus O_B, I_A I_B} & O'_A = 0 \wedge O'_B = 0 \\ \frac{1}{2}\delta_{O_A \oplus O_B, I_A (I_B \oplus 1)} & O'_A = 0 \wedge O'_B = 1 \\ \frac{1}{2}\delta_{O_A \oplus O_B, I_A \oplus 1} & O'_A = 1 \wedge O'_B = 0 \\ \frac{1}{2}\delta_{O_A \oplus O_B, I_A I_B \oplus 1} & O'_A = 1 \wedge O'_B = 1 \end{cases}. \quad (\text{B54})$$

The local operations are

$$T_{\text{LGYNl}}^{A|X} = \delta_{I_A, X} \delta_{O'_A, O_A} \delta_{A, X O_A} \quad (\text{B55})$$

$$T_{\text{LGYNl}}^{B|Y} = \delta_{I_B, Y} \delta_{O'_B, O_B} \delta_{B, Y O_B}. \quad (\text{B56})$$

Together, they yield the correlations

$$\overline{P}_{AB|XY}^{(\text{LGYNl}=11/12)} = \frac{1}{3}\delta_{A,0}\delta_{B,0} + \frac{2}{3}\delta_{A,XY}\delta_{B,XY}. \quad (\text{B57})$$

OCB = 1

For the OCB inequality, the process is just a relabeling of the PR box for fixed O'_A, O'_B :

$$W_{\text{OCB}} = \frac{1}{2} \delta_{O_A \oplus O_B \oplus O'_B, (I_A \oplus O'_A \oplus O'_B) I_B} , \quad (\text{B58})$$

and the local transformations have a particular simple form:

$$T_{\text{OCB}}^{A|X} = \delta_{I_A, 0} \delta_{O'_A, O_A \oplus X} \delta_{A, O_A} \quad (\text{B59})$$

$$T_{\text{OCB}}^{B|Y Y'} = \delta_{I_B, Y'} \delta_{O'_B, O_B \oplus Y} \delta_{B, O_B} . \quad (\text{B60})$$

The resulting correlations from this boxworld process and local operations is given in the form of Table III.

$(a, b) \setminus (x, y, y')$	(0, 0, 0)	(0, 0, 1)	(0, 1, 0)	(0, 1, 1)	(1, 0, 0)	(1, 0, 1)	(1, 1, 0)	(1, 1, 1)
(0,0)	1/2	1/2	0	1/2	1/2	0	0	0
(0,1)	1/2	0	0	0	1/2	1/2	0	1/2
(1,0)	0	1/2	1/2	1/2	0	0	1/2	0
(1,1)	0	0	1/2	0	0	1/2	1/2	1/2

Table III. Values for $\overline{P}_{AB|XY Y'}^{(\text{OCB}=1)}$ for the distribution saturating the OCB inequality. The first column and row indicate the values of (a, b) and (x, y, y') , respectively. The entries of the table are the value of the corresponding probability.